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BRIGHT ANODIZING BRIGHT

PROCESS VARIANT **BRIGHT**

= TYPE II OVER A BRIGHTENED SURFACE

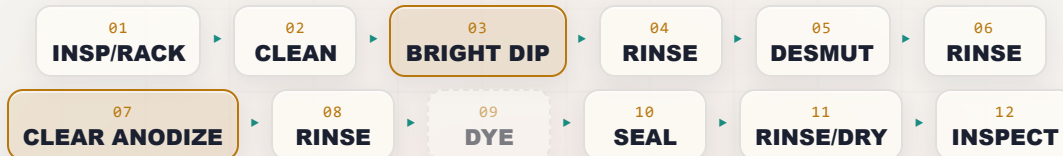
SPECULAR • DECORATIVE

MIRROR / SPECULAR DECORATIVE FINISH

CHEMICAL BRIGHT DIP + CLEAR SULFURIC ANODIZE • HIGH-PURITY ALUMINUM • DECORATIVE

THE COMPLETE TRAINING MANUAL

*A full training course for the brand-new operator, all the way to a signed certificate of completion. The decorative anodize whose whole product is **brilliance**: we polish the aluminum to a mirror in a hot acid **bright dip**, then grow a **clear, transparent anodize** to lock that shine in forever. The catch: that bright dip is **hot phosphoric-nitric acid that breathes out toxic brown NOx fumes** — so we teach the chemistry and the discipline.*



OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 • 2026 • FOR-DUMMIES STYLE

Training reference only. Always verify exact parameters against your chemistry supplier's Technical Data Sheets (TDS) — bright-dip baths are usually **proprietary** — customer/architectural/automotive specs (e.g., ASTM B580; MIL-PRF-8625 Type II for the clear anodize), and current SDS, and comply with OSHA, EPA, REACH, and local regulations before production use. **The hot phosphoric-nitric bright dip causes severe acid burns and evolves toxic brown NOx fumes — whose lung effects can be delayed for hours — and demands strong local exhaust and fume scrubbing. Sulfuric acid still causes severe burns; seal tanks run near boiling.**

— FIND YOUR WAY

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REMEMBER

This manual is a training and reference tool — **not** a substitute for your facility's written procedures, your supplier's TDS/SDS, your customer's spec (ASTM B580, MIL-PRF-8625, architectural/automotive standards), or your supervisor's instructions. When this manual and your shop's official procedure disagree, **follow your shop's official procedure and ask your supervisor.**

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU'RE IN THE RIGHT PLACE — HERE THE WHOLE POINT IS TURNING ALUMINUM INTO A MIRROR AND MAKING THE SHINE PERMANENT

Welcome. If you are holding this manual, you are about to learn how to run — or at least understand — a **bright anodizing line**. Maybe you started this week; maybe you've racked parts for a year and finally want to know *why*. Either way, you're in the right place.

Here is the first big thing to wrap your head around, and it's going to feel backwards if you've ever seen electroplating: **in anodizing, we are not coating your part with a different metal. We are growing a tough, glass-hard oxide layer out of the aluminum itself.** And to do that, your part is hooked up as the **anode** — the *positive* electrode — the exact opposite of plating. That's the single biggest *process* idea in this whole book.

Here is the second big idea, and it's the reason this process exists at all: **bright anodizing is the finish where the brilliance is the product.** Most decorative anodize you'll meet (standard satin Type II) is deliberately made *matte* — it's etched to a soft frosty texture. Bright anodizing does the opposite. We **polish the aluminum to a specular mirror first** — usually in a hot acid **chemical bright dip** — and *then* grow a **thin, perfectly clear anodize** over it so the mirror shows right through the coating and is locked in for good. Reflectors, automotive and appliance trim, cosmetics packaging, jewelry, lighting — anywhere a customer wants aluminum that gleams like chrome but won't rust or peel.



REMEMBER

Two stages make a part "bright": **(1) brighten** the surface to a mirror (the chemical bright dip — the heart of this manual), and **(2) clear-anodize** it thin and transparent so the shine shows through and is sealed in. Skip stage 1 and you have ordinary anodize; make stage 2 too thick or hazy and you bury the shine.

Do not read "decorative" as "harmless" either: the clear anodize is strong sulfuric acid, and the **bright dip is the most dangerous tank in this whole book** — hot (~90–110 °C) concentrated phosphoric + nitric acid that causes instant severe burns *and* breathes out **toxic brown NOx fumes** whose worst lung effects can be **delayed for hours**.

The third thing up front: **this manual assumes you know nothing about anodizing, chemistry, or electricity.** That's a promise, not an insult — every term is defined the first time we use it, and every step explains not just *what* but *why*. Don't memorize it; *understand* it.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: clean (gently — **no matte etch**), **bright dip**, rinse, desmut, rinse, **clear-anodize**, rinse, (rarely dye), **seal clear**, rinse/dry, inspect. Reading it in order matters: the order of an anodizing line is not random — and neither is this book's.



TIP

Read the Fundamentals (Part One) through *before* any station chapter — **especially Chapter 3 (the brighten-then-anodize concept) and Chapter 7 (why bright demands high-purity aluminum)**. The station chapters assume you grasp the big picture: the part is the anode, oxide grows out of it, the coating is *thin and clear* on purpose, and the finish lives or dies on a mirror-smooth, high-purity surface.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS



TIP

A shortcut or trick of the trade — the kind of thing a 20-year veteran would lean over and tell you.



REMEMBER

A core, load-bearing idea. If you forget everything else on the page, don't forget these.



HEADS-UP

A safety warning. Read every single one — these protect your skin, eyes, **lungs**, and long-term health. On this line we use them *very* liberally.



TECHNICAL STUFF

Deeper chemistry for the curious. Optional — skip it and you can still run the line.



FUN FACT

A bit of trivia to make the science stick (and to give you something to say at lunch).

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

MASTER THIS AND EVERY STATION CHAPTER CLICKS INTO PLACE

By the time you finish this manual, you should be able to confidently:

1. **Explain what anodizing is** in plain language — and how it is fundamentally *different from electroplating*: the part is the **anode**, and we **grow an aluminum-oxide layer out of the aluminum itself** instead of depositing a different metal onto it.
2. **State what bright anodizing is and why it exists**: a two-stage decorative process — **brighten the aluminum to a specular mirror, then grow a clear (transparent) anodize over it** — so the brilliance becomes a permanent, integral, corrosion-resistant finish. The visual shine *is* the product.
3. **Describe the two brightening routes** — **chemical bright dip** (a hot phosphoric-nitric acid bath that chemically levels/smooths the surface, the dominant method) and **electrobrightening/electropolishing** (an electrochemical leveling alternative) — and recognize the bright dip as the heart of this line.
4. **Explain why the anodize here is thin and clear** — a transparent oxide on high-purity aluminum lets the mirror show through; **too thick or too hazy a coating dulls the very brilliance you created**.
5. **Explain why bright anodizing demands high-purity aluminum** (super-purity 99.99% or special bright alloys like **5657, 5252, 6463**) — iron, silicon, and copper form light-scattering particles that haze and dull the finish; **casting and high-Si/high-Cu alloys cannot go truly bright**.
6. **Explain why bright anodize deliberately skips the matte caustic etch** — etching frosts the surface, which is the *opposite* of what we want — and contrast it with standard satin Type II.
7. **Explain the porous oxide and dimensional growth** — anodic oxide is porous (so it can be sealed), and grows roughly half-in/half-out; because bright anodize is *thin*, the change is small.
8. **Place bright anodize in the family** — Type I (chromic), Type II (sulfuric), Type III (hardcoat), plus specialty variants — and recognize that **bright anodize is fundamentally a thin, clear sulfuric Type II anodize over a brightened surface**, governed by standards such as **ASTM B580**.
9. **Recognize the main hazards** — above all the **hot phosphoric-nitric bright dip with its severe burns and toxic, delayed-effect NOx fumes**, plus sulfuric acid, hydrogen at the cathode, DC electrical/arcing, and the near-boiling seal — and know the correct PPE, ventilation, emergency response, and waste handling.
10. **Use the field checks** a bright anodizer relies on — **specular gloss (gloss meter), distinctness-of-image (DOI), water-break, coating thickness, and seal-quality tests** — and **talk the language** with your lead, chemist, and supplier.

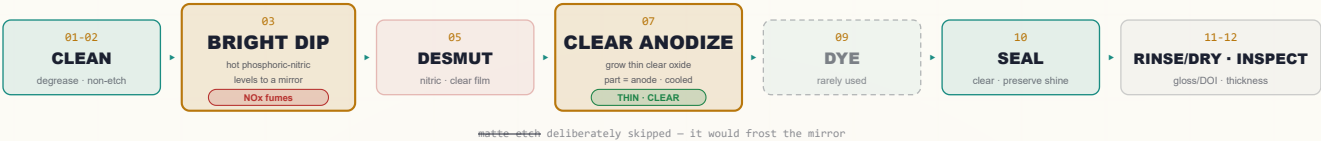


REMEMBER

You are not expected to hit all of these on day one. Bright anodizing is a craft, and a finicky one — brilliance is unforgiving of small mistakes. But the *safety* objectives (#9) are not optional and not gradual. You must have those — **especially the bright-dip NOx and burn hazards** — before you work the line at all.

• THE LINE AT A GLANCE

Don't memorize yet — feel the rhythm: **Inspect/Rack** → **Clean (no matte etch)** → **BRIGHT DIP** → **Rinse** → **Desmut** → **Rinse** → **CLEAR ANODIZE (thin)** → **Rinse** → **(Dye, rarely)** → **Seal (clear)** → **Rinse/Dry** → **Inspect**.



FUN FACT

The aluminum oxide we grow is chemically the same family as **sapphire** (crystalline aluminum oxide, Al_2O_3) — and on a bright-dipped, high-purity surface that oxide is so **clear** that you look straight through it to the mirror underneath. You're not painting on shine; you're sealing a polished metal mirror under a layer of transparent ceramic.

CHAPTER 1

WHAT IS ANODIZING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT ANODIZING IS

THE ONE-SENTENCE VERSION

Anodizing is using electricity to grow a hard, protective oxide layer out of the surface of an aluminum part — the part itself becomes part of the coating. That's the whole idea. Now let's unpack it, because it's genuinely different from plating and that difference trips up almost everyone at first.

FIRST, WHAT PLATING DOES (SO YOU CAN SEE THE DIFFERENCE)

In electroplating — zinc, nickel, chrome — you take your part, hook it up as the **cathode** (the *negative* electrode), and you *deposit a different metal* onto it from a bath. The coating is a *separate metal sitting on top of* your part.

Anodizing flips almost all of that:

- Your part is the **anode** — the **positive** electrode. (That's literally where the word “anodize” comes from.)
- We don't add a different metal. We take the **aluminum that's already there** and convert its surface into **aluminum oxide** by feeding electricity through an acid bath.
- The coating isn't sitting *on* the part — it **is** the part's surface, chemically grown and locked in. It can't peel like paint or flake like a bad plate, because it's integral to the metal.



REMEMBER — THE SINGLE BIGGEST IDEA IN THIS BOOK

In plating, the part is the **cathode** and you **add** a metal. In anodizing, the part is the **anode** and you **grow oxide out of the part itself**. If you remember nothing else from Chapter 1, remember this. Everything else follows from it.

HOW IT PHYSICALLY WORKS: THE KITCHEN-SINK PICTURE

Picture a tank full of liquid called the **electrolyte** (or just “the bath”). For the clear anodize on this line, that liquid is **dilute sulfuric acid**. You hook two things to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction “DC” current anodizing needs):

- **The anode** — this is your **aluminum part**, connected to the **positive** terminal. “*Anode = the part, the thing growing oxide.*”
- **The cathode** — bars or plates (commonly **lead or aluminum 6063**) hanging in the same tank, connected to the **negative** terminal. The cathode just completes the circuit and is where **hydrogen gas** bubbles off.

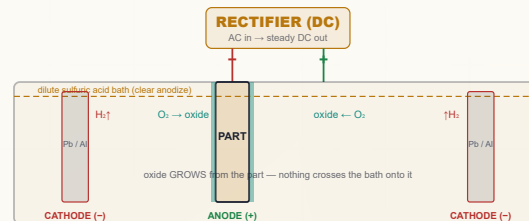


TECHNICAL STUFF — THE POROUS STRUCTURE, IN BRIEF

The sulfuric acid is slightly aggressive, so as oxide forms, the acid is *also* gently dissolving it. That tug-of-war builds a layer of microscopic vertical **pores** — which is why anodize can later be dyed and *must* be sealed (Chapter 4). On a bright line we keep this oxide **thin and clear** so it acts as an invisible window over the mirror.

When you turn on the rectifier, here's the dance that happens:

1. Current flows. At the surface of your aluminum part (the anode), water in the bath is split and **oxygen** is delivered right at the metal surface.
2. That oxygen immediately combines with the surface aluminum to form **aluminum oxide (Al₂O₃)** — a hard, ceramic layer.
3. The acid is slightly aggressive, so as the oxide forms, the acid is *also* gently dissolving it — and this tug-of-war creates the famous **porous structure** (Chapter 4).
4. Over at the cathode, **hydrogen gas** bubbles off into the air (a safety item — flammable gas — we'll come back to it).



Compare to a plating cell: the labels are reversed — here the **part is the ANODE (+)** and the counter-electrodes are the cathodes.



TECHNICAL STUFF — THE TWO REACTIONS

At the part (anode), aluminum is oxidized: simplified, $2\text{Al} + 3\text{H}_2\text{O} \rightarrow \text{Al}_2\text{O}_3 + 6\text{H}^+ + 6\text{e}^-$. At the cathode, those electrons reduce hydrogen ions: $6\text{H}^+ + 6\text{e}^- \rightarrow 3\text{H}_2\uparrow$. So **oxygen builds the coating at your part; hydrogen gas leaves at the counter-electrode**. No aluminum travels through the bath onto something else — it stays put and turns to oxide in place.



TECHNICAL STUFF — WHY THE BATH MUST BE COOLED

The reaction is **exothermic** and a lot of electrical energy turns to heat right at the part surface. If the bath gets too warm, the acid dissolves the oxide too aggressively — you get a soft, powdery, “burned,” **hazy** coating instead of a hard, clear one. On a *bright* line that haze is fatal to the finish. So the clear-anodize tank has **refrigerated cooling and air agitation** to hold it near ~20 °C (68 °F). On this line, *cold and well-stirred is brilliance*.



FUN FACT

Anodizing is one of the few finishing processes where the coating is *grown from* the part rather than *added to* it. That's why anodized aluminum can't “chip off” to bare metal the way paint or a bad plate can — there's no glue line. The oxide and the metal are continuous, which is exactly why a bright-anodized mirror finish stays bright: the shine is sealed *inside* the metal's own skin.

CHAPTER 2

WHAT IS BRIGHT ANODIZING?

THE FINISH YOU CAN SEE YOUR FACE IN — BUILT FOR BRILLIANCE, NOT JUST PROTECTION

Bright anodizing is producing a **brilliant, specular (mirror) aluminum finish and then anodizing to lock it in**. Unlike standard satin Type II — which is intentionally etched to a soft matte look — bright anodizing keeps (or creates) a **high-gloss, reflective surface** and protects it under a **thin, transparent anodic oxide**. The visual brilliance *is* the deliverable.

It's a two-stage idea (Chapter 3 goes deep):

1. **Brighten** — chemically (a hot phosphoric-nitric **bright dip**) or electrolytically (**electrobrightening/electropolishing**) — to level the surface to a mirror.
2. **Clear-anodize** — a thin, transparent sulfuric (Type II) anodize, then **seal** — so the mirror shows through and is permanently protected.

• WHY SHOPS CHOOSE IT

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
Permanent mirror shine	The brilliance is sealed under integral oxide — it won't tarnish, oxidize dull, or fingerprint-etch like bare polished aluminum
Corrosion + wear protection	The clear oxide protects the polished surface far better than bare metal, especially once sealed
Integral & non-flaking	Grown from the metal — it won't peel or chip to bare aluminum like chrome plate or paint can
Decorative without plating	A bright, chrome-like look without electroplating a different metal (no nickel/chrome bath, no Cr VI)
Lightweight & recyclable	All the shine of polished metal at aluminum's weight
Can be lightly tinted	A thin clear coat can take a light, transparent dye (e.g., gold/bronze "anodized aluminum" trim)

WHERE YOU'LL SEE IT

Reflectors and lighting (the classic application — high-purity bright-anodized reflector sheet), **automotive and appliance trim**, **cosmetics and perfume packaging** (lipstick cases, compacts, caps), **jewelry and watch components**, **decorative architectural accents** (bright fascia, name plates, signage), and premium consumer-product trim.

★

REMEMBER

Bright anodize's superpower is **specular brilliance made permanent**. Two things must both be true to deliver it: the surface must start (or be made) **mirror-smooth**, and the anodize over it must stay **thin and clear**. Lose either one and you've made an ordinary, dull anodize — which on a bright job is scrap.

💡

TIP

When a print says just "anodize," it almost always means **satin Type II**. "Bright anodize," "bright dip," "specular," "mirror finish," "reflector finish," or a **gloss/DOI** callout means *this* process. Check the spec and ask — because the surface prep is completely different (bright dip vs. matte etch) and the alloy requirement is far stricter.

☀️

FUN FACT

The shiny aluminum reflector behind the bulb in many flashlights, headlamps, and light fixtures is **bright-anodized super-purity aluminum**. It's chosen over a mirror or chrome plating because it reflects efficiently, weighs almost nothing, won't rust, and the clear anodize keeps it bright for the life of the product.

BRIGHTEN FIRST, THEN LOCK IT IN

THE TWO-STAGE HEART OF THE WHOLE MANUAL — THE BRIGHT DIP IS WHERE THE MAGIC (AND THE BIGGEST HAZARD) LIVES

This is the chapter that makes bright anodizing *bright*. Slow down here — and read the safety notes as carefully as the chemistry.

Standard anodize starts with a **matte caustic etch** that frosts the surface. Bright anodize does the opposite: it **levels the surface to a mirror first**, then grows a clear coat. There are two ways to brighten. The dominant one — and the heart of this line — is the **chemical bright dip**.

ROUTE 1 — CHEMICAL BRIGHT DIP (CHEMICAL BRIGHTENING / CHEMICAL POLISHING)

A **hot, concentrated acid bath** that *chemically levels* the surface. The classic chemistry is **phosphoric acid + nitric acid** (sometimes phosphoric-sulfuric, or a three-acid phosphoric-sulfuric-nitric blend), usually as a **proprietary product** (e.g., “Alupol/Phosbrite-type” baths) with surfactants and trace additives.

- **How it makes a mirror:** the bath **dissolves microscopic peaks faster than valleys**. A thick, viscous phosphoric “diffusion layer” sits over the surface; the nitric acid is the **oxidizer** that drives the dissolution; the high points stick up into fresher acid and dissolve preferentially. Net effect: the surface **levels and smooths** to a specular, reflective finish — *chemical polishing*, no buffing wheel required.
- **Typical conditions:** **~90–110 °C** (commonly ~95–105 °C), **short immersion ~30 s to ~3 min**, high acid concentration (phosphoric is the bulk; nitric a few percent).
- **The byproduct that defines the safety of this manual:** as nitric acid does its work, it gives off **NOx — brown/orange nitrogen-oxide fumes (NO, NO₂)** — a serious **inhalation hazard**.



HEADS-UP — THE STANDOUT HAZARD OF THIS ENTIRE MANUAL

The bright dip is **hot (~90–110 °C) concentrated phosphoric + nitric acid**. It causes **instant, severe burns** on contact, and it **breathes out toxic brown NOx fumes**. NOx is a deep-lung irritant whose worst effect — **pulmonary edema (fluid in the lungs)** — can be **delayed for hours** after an exposure that felt minor at the time. This tank **must** have strong **local exhaust / fume scrubbing**, and you must never work it without ventilation confirmed running. This is not a “be careful” tank. This is a “respect it like your life depends on it” tank — because it does.



TECHNICAL STUFF — WHY PHOSPHORIC AND NITRIC

Phosphoric acid is **viscous** and forms the diffusion-controlled film that produces *leveling* (smoothing of microroughness). Nitric acid is the **oxidizer/depolarizer** that keeps metal dissolving and gives **brightening** (specular reflectivity). Together they level *and* brighten. The nitric is also exactly what generates the NOx fumes — so the bright dip’s two best features (the shine) and its worst hazard (the fumes) come from the **same** ingredient. Three-acid baths use less nitric and fume somewhat less, trading a little brightness for lower NOx.

ROUTE 2 — ELECTROBRIGHTENING / ELECTROPOLISHING

An **electrochemical** alternative: the part is made anodic in a special electrolyte and the surface is leveled by **controlled anodic dissolution** (high points dissolve fastest). For aluminum this includes alkaline electrobrightening (e.g., the historic **Brytal** carbonate/phosphate process) and phosphoric-sulfuric electropolishing. It can give excellent brilliance and avoids the NO_x of nitric bright dips, but needs DC, careful control, and its own electrolyte hazards. Many shops favor the **chemical bright dip** for speed and simplicity — which is why this manual centers on it.

THEN: THE CLEAR ANODIZE LOCKS IT IN

Once the surface is a mirror, we grow a **thin, transparent sulfuric (Type II) anodize** over it and **seal** it. On **high-purity aluminum** that oxide is so clear you see straight through to the polished metal — and now the shine is **integral, sealed, and protected**. (Why thin? Chapter 4. Why high-purity? Chapter 7.)



REMEMBER — THE HEART OF BRIGHT ANODIZING

Brighten → **Clear-Anodize** → **Seal**. Brighten levels the surface to a mirror. Clear-anodize protects it with a *transparent* coat the shine shows through. Seal closes the pores to keep it brilliant and corrosion-resistant. This “brighten-first” idea is what separates this line from every other anodize — and the bright dip is both its soul and its single biggest hazard.



TIP

The bright dip can only *level what's already fairly smooth* — it removes micro-roughness, not deep scratches, gouges, or coarse machining marks. Parts usually arrive **mechanically polished/buffed or as bright-rolled sheet** first; the bright dip then takes them from “shiny” to “mirror.” Garbage in, garbage out: a scratched part comes out a *shiny scratched part*.



FUN FACT

A good chemical bright dip can take a buffed aluminum part to over **80–90 gloss units** at 60° — bright enough to read reflected text in. The whole trick is microscopic: the acid is shaving down hills you can't even see, one tenth of a micron at a time, until light reflects like off still water.



TECHNICAL STUFF — “BRIGHT DIPPED ONLY” VS. ANODIZED

Some parts are sold **bright-dipped only** (no anodize) for indoor decorative use — but without the clear anodize the shine isn't protected and will slowly dull, tarnish, or fingerprint-etch. The anodize is what makes the brilliance *last*. On this line, the bright dip *creates* the mirror and the clear anodize *preserves* it — two different jobs, back to back.

THE CLEAR OXIDE

TRANSPARENT SO THE SHINE SHOWS THROUGH — THE SAME POROUS OXIDE AS ANY ANODIZE, KEPT THIN AND CRYSTAL-CLEAR ON PURPOSE

The anodic oxide on this line works exactly like any other anodize at the microscopic level — and then we deliberately keep it **thin and transparent**.

THE OXIDE GROWS POROUS (AND WHY THAT'S STILL TRUE HERE)

Remember the tug-of-war from Chapter 1: oxide forms while the sulfuric acid gently dissolves it. The result is a layer of **billions of tiny columns with a microscopic pore down the center of each** — a honeycomb of vertical pores — sitting on a thin, dense **barrier layer** against the metal. Those pores are why anodize can be **dyed and must be sealed** (just like Type II).



BUT ON A BRIGHT LINE, TWO THINGS ARE DIFFERENT

1. It's kept THIN. A thicker oxide **scatters more light** and slightly reduces gloss and distinctness-of-image. So bright/decorative anodize runs on the **thin** side — typically **~3–8 μm** — versus the heavier satin Type II. Thinner = brighter; thicker = more protective but duller. **2. It's kept CLEAR.** On high-purity aluminum, the oxide is **transparent**, so the polished mirror underneath shows right through it. On the wrong (impure) alloy, the same oxide comes out **hazy/gray** and the brilliance is lost (Chapter 7).

★ REMEMBER

On this line, the oxide's job is to be an **invisible, protective window** over a mirror — not a thick armor coat. **Thin and clear preserves brilliance; thick and hazy destroys it.** That single trade-off shapes how the anodize tank is run. Dye is usually **skipped** (or a very light transparent tint), because heavy color muddies the shine; then we **seal** with chemistry chosen to stay clear and bloom-free.



HEADS-UP — SEALING IS STILL A ONE-WAY DOOR

Once you seal, you **cannot dye** — the pores are shut. If a light tint is specified, it goes **before** seal. And a **wrong seal** (hard water, wrong pH) leaves a chalky **bloom** that ruins a mirror finish — so seal quality matters even more on a bright job than on a satin one.

CHAPTER 5

DIMENSIONAL GROWTH

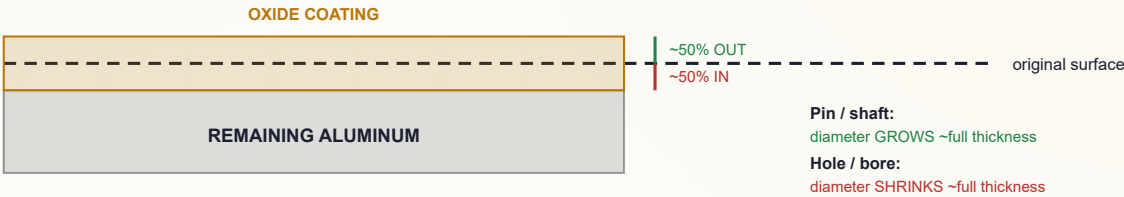
THE COATING MAKES THE PART BIGGER — BUT BECAUSE BRIGHT ANODIZE IS THIN, THE NUMBERS ARE SMALL

Because we convert aluminum into aluminum oxide, and oxide takes up **more room** than the metal it came from, the coating grows in **two directions at once**:

- About **half the coating thickness grows *into*** the original surface (consuming aluminum).
- About **half grows *out*** from the original surface (the part gets bigger).

For a thin bright-anodize coating — say **5 μm** — each surface moves out only **$\sim 2.5 \mu\text{m}$** , and a diameter changes by about the **full $\sim 5 \mu\text{m}$** . That's small, which is *good* for decorative trim with fine detail — but it's not zero, and **the bright dip *also* removes metal** before you even anodize.

Bright: $\sim 3\text{--}8 \mu\text{m}$ $\rightarrow$ change is very small



REMEMBER — COUNT THE BRIGHT DIP, TOO

On this line, dimension changes happen **twice**: the **bright dip removes a little metal** (it's dissolving the surface to level it — typically a fraction of a μm to a couple μm), and then the **anodize grows $\sim$ half-in/half-out**. For tight-tolerance bright parts, plan for *both* — the etch you skipped is replaced by a (gentler) bright-dip metal loss.



TIP

Because bright dip + thin clear anodize remove/grow only small amounts, bright anodize is friendly to **fine decorative detail and crisp edges** — one reason it suits trim, logos, and jewelry. Still confirm critical fits against the print.



HEADS-UP — MASKING AND CONTACTS

Areas that must *not* grow or change (threads, bearing fits, electrical-ground points) are usually **masked**. And remember anodize is **electrically insulating** — any surface that must conduct after finishing has to be masked or it won't work. Masking on bright parts must also be clean-edged so it doesn't leave a witness line on a mirror.



TECHNICAL STUFF — THE PILLING-BEDWORTH IDEA

Aluminum oxide occupies more volume than the aluminum it formed from (Pilling–Bedworth ratio > 1). That expansion is *why* the coating grows outward at all — and the same expansion lets hot-water sealing pinch the pores shut. One fact (oxide is bulkier than metal) explains both growth and sealing.

CHAPTER 6

WHERE BRIGHT ANODIZE FITS

SAME CORE IDEA, SEVERAL FLAVORS — BRIGHT ANODIZE IS A THIN, CLEAR SULFURIC ANODIZE OVER A MIRROR

All anodize types grow an integral aluminum-oxide coating with the part as the anode. They differ in **acid, thickness, and purpose**.

	TYPE I — CHROMIC	TYPE II — SULFURIC	TYPE III — HARDCOAT	BRIGHT (THIS MANUAL)
Electrolyte	Chromic acid	Sulfuric acid	Sulfuric (colder/stronger)	Sulfuric (Type II), thin
Surface prep	Etch varies	Matte caustic etch	Light/none	Bright dip (no matte etch)
Thickness	~0.5–7.5 μm	~5–25 μm	~25–100+ μm	~3–8 μm (kept thin)
Look	Gray	Satin, dyeable	Dark, hard	Specular mirror, clear
Main use	Aerospace fatigue	Decorative + protective	Max wear	Decorative brilliance

**REMEMBER**

Bright anodize is not a separate “Type.” It is, at heart, a **thin, clear sulfuric Type II anodize** — with two defining twists: the surface is **brightened (not matte-etched)** beforehand, and **high-purity aluminum** is required. Everything you’d learn about Type II applies; bright anodize just adds the brighten-first step and demands a better substrate.

**TIP**

Other specialty anodizes in this training series: **BSAA** (boric-sulfuric, a chromic replacement for aerospace), **PAA** (phosphoric-acid, for adhesive bonding), **hardcoat (Type III)**, and the **coloring methods** (organic dye, two-step electrolytic, integral color). Bright anodize is the **decorative/specular** member. Don’t confuse the bright-dip *pretreatment* (chemical polishing) with *PAA anodizing* — both use phosphoric acid but do entirely different jobs.

**TECHNICAL STUFF — PRETREATMENT VS. COATING, IN SEQUENCE**

A subtle but important point: the **bright dip is a chemical pretreatment** (it dissolves/levels metal, no current), while the **clear anodize is an electrochemical coating step** (current on, oxide grown). A bright-anodize line therefore has *two* surface-chemistry stages doing very different things back-to-back. Refs note: governed by anodic-coating standards such as **ASTM B580** (architectural/decorative classes), with the clear-anodize portion often referenced to **MIL-PRF-8625 Type II**; the bright-dip chemistry itself is typically **proprietary** (follow the supplier TDS).

CHAPTER 7

ALUMINUM PURITY & ALLOYS

THE DEFINING CONSTRAINT: YOU CANNOT MAKE A MIRROR OUT OF DIRTY METAL

Anodizing only works on **aluminum** (and a few other “valve metals”). But for **bright** anodizing, “aluminum” isn’t enough — it has to be the **right, high-purity aluminum**. This is the single most important material rule in the manual, and it’s where most bright-anodize disappointments come from.

WHY PURITY DECIDES BRILLIANCE

Brilliance is about **light reflecting cleanly off a smooth, uniform surface**. Alloying elements and impurities — especially **iron (Fe), silicon (Si), copper (Cu), and manganese (Mn)** — form tiny hard **second-phase / intermetallic particles** scattered through the metal. These particles **don’t dissolve or anodize uniformly** with the surrounding aluminum, leaving **microscopic pits, specks, and a frosty haze**, and they **scatter light** so the clear oxide looks **gray, cloudy, or milky**. The purer the aluminum, the fewer of these particles, the brighter the result.

MATERIAL	BRIGHTNESS BEHAVIOR
99.99% / 99.9% super-purity Al	Best — true mirror (reflectors, jewelry, cosmetics)
5657 / 5252 / 5357 (bright sheet)	Excellent — designed for bright trim
6463 (bright extrusion)	Excellent — bright architectural/automotive trim
6063 (architectural)	Good — bright, but a step below 6463
5xxx general / 6061	Fair — anodize fine but not top brilliance
2xxx (copper)	Poor — dull / yellow
Cast / high-silicon	Cannot go bright — gray / mottled

**REMEMBER — THE DEFINING CONSTRAINT**

No mirror without high-purity aluminum. Iron, silicon, and copper form light-scattering particles that haze the finish; **castings and high-Si/high-Cu alloys cannot bright-anodize to a true mirror**. On a bright job the alloy and pedigree of the metal matter *as much as everything you do on the line*. Confirm the alloy **before** you promise brilliance.

**TIP**

Always verify the **alloy and temper on the traveler** before running a bright job, and when in doubt, **bright-dip and anodize a sample of the actual lot first**. A customer who supplies a casting and expects a mirror is going to be disappointed no matter how perfectly you run the line — and that conversation is far cheaper *before* you process the parts.

**TECHNICAL STUFF — CONSTITUENT PARTICLES AND “FROST”**

The Fe/Si/Cu/Mn “constituent particles” are typically a few microns across and are either cathodic or anodic relative to the aluminum matrix. In the bright dip and anodize they’re attacked at different rates than the matrix, leaving **micro-pits and a diffuse-reflecting (frosted) micro-texture**. That’s why two parts that look identical bare can finish completely differently: the one with more/larger constituent particles “frosts” while the high-purity one stays specular.

**FUN FACT**

Lighting and optics makers buy **super-purity (99.99%) aluminum reflector sheet** precisely for this reason: every fraction of a percent of impurity costs reflectivity. Bright anodizing on 4N aluminum can reach **total reflectance in the 80s–90s of percent** — competitive with silvered mirrors, but lighter, tougher, and corrosion-protected by the clear oxide.

CHAPTER 8

HOW THE LINE WORKS

WALKING THE LINE — EVERY TANK HAS A JOB, AND THE ORDER IS THE RECIPE

A bright anodize line is a sequence of tanks. Here's the whole journey and *why each step has to be where it is*.

1. **Inspect / Rack** — Confirm **alloy/purity** (mirror only happens on the right metal) and spec; rack for firm, current-carrying contact. The part **is the anode**, so a loose contact = no coating or a burned spot.
2. **Clean / Degrease (non-etching)** — Remove oils and soil with a **mild, non-etching** cleaner. We must **not** frost the surface — so no aggressive caustic etch here.
3. **BRIGHT DIP (hot phosphoric-nitric)** — The heart. Chemically **level the surface to a mirror**. (This *replaces* the matte etch of a satin line.) Severe burns + **toxic NOx fumes** — ventilation mandatory.
4. **Rinse** — Wash off hot phosphoric-nitric drag-out before it contaminates downstream tanks.
5. **Desmut / Deox (nitric)** — The bright dip leaves a thin **film/smut**; a nitric dip clears it to a bright, uniform, active surface.
6. **Rinse** — Clean handoff before the main tank.
7. **CLEAR ANODIZE** — Part = anode, dilute sulfuric, cooled, DC on. Grow a **thin, transparent** porous oxide.
8. **Rinse (cool DI)** — Gently rinse acid from the open pores without pre-sealing.
9. **Dye (rarely)** — Only if a light transparent tint is specified; most bright work skips this.
10. **Seal (clear)** — Close the pores to lock in brilliance and corrosion resistance, **without bloom**.
11. **Rinse / Dry** — Final clean DI rinse and gentle, spot-free dry.
12. **Inspect** — **Gloss/DOI**, thickness, seal quality, appearance.



REMEMBER — WHY THE ORDER CAN'T MOVE

You can't bright-dip *after* anodizing (the oxide would block leveling and you'd ruin the coating). You can't anodize over bright-dip smut (hazy, streaky oxide). You can't seal before a tint (sealed pores won't take color). And you must **never add a matte etch** — it frosts the mirror you're trying to create. The line *is* the recipe.



HEADS-UP — THE LINE IS A TOUR OF HAZARDS, AND ONE IS EXCEPTIONAL

Walking the line you pass a **mild alkaline cleaner**, the **hot phosphoric-nitric bright dip** (**severe burns + toxic NOx fumes — the standout**), **nitric desmut**, **strong sulfuric** (anodize), **flammable hydrogen** (cathode), **high-current DC**, and **near-boiling seal water**. Station 03 demands the most respect of any tank in this book.



TIP

When you're learning, walk the empty line with your trainer and name each tank's *job* and *main hazard* out loud. If you can do that for all twelve — and especially if you can recite the bright dip's NOx hazard from memory — the rest of the manual is mostly detail.

CHAPTER 9

PPE — YOUR DAILY ARMOR

WHAT YOU WEAR, AND WHY EACH PIECE IS THERE

A bright anodize line works with a **hot concentrated phosphoric-nitric bright dip (plus its toxic fumes)**, nitric desmut, strong sulfuric anodize, near-boiling seal tanks, high-current DC, and chemical mist. None of these will hurt you if you respect them and wear the right gear. All of them can hurt you badly if you don't — and the bright dip can hurt your **lungs** as well as your skin.

HAZARD ON THE LINE	PPE / CONTROL THAT PROTECTS YOU
Hot phosphoric-nitric bright dip (~90–110 °C) — severe burns, splash	Sealed chemical splash goggles + face shield , heat- and acid-resistant gloves (elbow length), chemical apron/suit, chemical boots
NOx (brown) fumes from the bright dip — toxic inhalation, delayed lung effects	Local exhaust / fume scrubbing FIRST ; respiratory protection per your program (e.g., acid-gas cartridges) if exposure possible; never work it without ventilation confirmed
Nitric desmut — acid burns, may also evolve NOx	Same acid PPE; ventilation
Sulfuric anodize — burns, splash, acid mist	Sealed goggles, face shield, gloves, apron, boots; local exhaust
Near-boiling seal (~96–100 °C) — scald/steam	Face shield, heat gloves, long sleeves; mind the steam
DC electrical / arcing at contacts	Dry hands/gloves, no jewelry, follow LOTO
Slips on wet floors	Non-slip chemical-resistant boots

 **TIP**

Put your gear on *in order*: boots and apron first, then gloves, then goggles, then face shield last. Take it off in reverse, and **rinse your gloves before removing them**. This keeps you from touching your face with contaminated gloves — important everywhere, critical near the bright dip.

 **REMEMBER**

On this line your **eyes, skin, AND lungs** are all exposed. Strong hot acid, near-boiling water, *and* a tank that breathes toxic gas all live here. **Sealed splash goggles, a face shield, and working ventilation are not optional** at the bright dip.

 **HEADS-UP — GOGGLES, NOT JUST SAFETY GLASSES**

Safety glasses stop flying chips; they do **not** stop a hot-acid splash from getting into your eyes from the side. At every chemical tank — above all the bright dip — sealed splash goggles **and** a face shield (when charging, dumping, or handling parts) are required.

 **HEADS-UP — NEVER EAT, DRINK, SMOKE, VAPE, OR APPLY COSMETICS ON THE LINE**

Hand-to-mouth contact is a major route for ingesting chemicals. Wash thoroughly before breaks and before leaving. And smoking/open flame is doubly forbidden here — hydrogen at the cathode is flammable.

CHAPTER 10

EMERGENCY & FIRST-RESPONSE

KNOW THESE COLD BEFORE YOU TOUCH A TANK — ESPECIALLY THE NOX RESPONSE

In an emergency, seconds matter and you won't have time to read.

SITUATION	WHAT TO DO — IMMEDIATELY
Acid on skin (bright dip, nitric, sulfuric)	Flush immediately with water at least 15 minutes ; remove contaminated clothing while flushing. Bright-dip acid is hot — treat as a chemical and thermal burn. Get medical attention.
Acid in eyes	Go straight to the eyewash , hold eyes open, flush at least 15 minutes , get medical help. Don't stop early.
Large splash / drench	Safety shower — full 15 minutes, clothing off.
Breathed NOx (brown fumes) from the bright dip	Get to fresh air immediately. Then get medical evaluation even if you feel “okay” — NOx lung injury (pulmonary edema) can be DELAYED for hours. Tell medical staff it was nitrogen-oxide exposure. Rest; exertion can worsen it. Do not “tough it out.”
Hot seal-tank scald	Cool with water; don't pop blisters; medical attention for anything beyond minor.
Acid spill	Alert others; contain only if trained; use the spill kit; follow the spill plan. Never mix acids and caustics during cleanup.
Electrical / arc at rectifier or contacts	Don't touch — de-energize via rectifier controls / LOTO. Never reach into a tank with current on.
Hydrogen gas	Keep ignition sources away; ensure ventilation is running.

HEADS-UP — NOX IS A DELAYED KILLER; DO NOT UNDERESTIMATE IT

Brown nitrogen-oxide fumes are a **deep-lung irritant**. A worker can breathe a dangerous dose, feel only mild throat/eye irritation, go home — and develop **life-threatening pulmonary edema hours later**. **Any** suspected NOx inhalation gets fresh air, rest, and a **medical evaluation with observation** — not “wait and see.” This is the most important first-aid lesson in the manual.

HEADS-UP — THE GOLDEN RULE FOR ACID MAKE-UP

Always add ACID to WATER, never water to acid. (“Do as you *oughta*: add acid to water.”) This applies to phosphoric, nitric, *and* sulfuric. Adding water to concentrated acid can flash-boil and erupt. Bath make-up is usually authorized-personnel work — but every operator must know this rule.

HEADS-UP — LOCKOUT/TAGOUT (LOTO)

LOTO is mandatory for *all* rectifier and electrical maintenance. Physically lock the power off and tag it. “I'll just be a second” is how people get electrocuted.

REMEMBER

Know where the **nearest eyewash, safety shower, spill kit, and fresh-air exit** are at every station before your shift starts. At the bright dip, also confirm the **exhaust/scrubber is running** before any work. In an emergency you won't have time to look.

CHAPTER 11

THE SAFETY PHILOSOPHY OF A BRIGHT LINE

THE MINDSET THAT TIES EVERY STATION TOGETHER

1. THE BRIGHT DIP IS THE MOST DANGEROUS TANK IN THE BOOK.

Hot concentrated phosphoric-nitric acid that burns on contact *and* breathes toxic, delayed-effect NO_x fumes. **Ventilation first, PPE always, respect total.**

2. ACID ALL THE WAY DOWN — BUT THE GAS IS THE SURPRISE.

Unlike a satin line (caustic etch *and* acid), a bright line is **mostly acidic**, so there's no high-pH/low-pH mixing to fear at the etch — but it adds a hazard most operators have never met: **toxic fume inhalation with delayed effects**. Learn to fear what you can't feel.

3. THE PART IS LIVE.

The part is the anode, so there's **DC current at the work and contacts**. Wet hands + current + acid is a bad combination.

4. HOT IS A HAZARD TOO.

The bright dip runs near boiling *and* the seal tank runs near boiling. Steam and near-boiling splashes scald.

5. HYDROGEN IS FLAMMABLE; ENGINEERING CONTROLS FIRST.

The anodize cathode bubbles off hydrogen — ventilation on, ignition sources away. Tank exhaust, **fume scrubbing**, cooling, lids, and splash guards do the heavy lifting; PPE is your last layer, not your first. And when in doubt, **stop and ask** — the unsafe operator is the one who guesses.

• THE HAZARD FAMILIES TO INTERNALIZE

1. **Toxic fume (the bright-dip headline).** NO_x from the phosphoric-nitric bath. *Defense:* local exhaust/scrubbing, never run without ventilation, evacuate to fresh air and **get medical evaluation** on any exposure (delayed effects).
2. **Corrosives — and they're HOT.** Hot phosphoric-nitric (~90–110 °C), nitric desmut, sulfuric anodize. *Defense:* goggles, face shield, gloves, apron; 15-minute flush; acid-to-water.
3. **Thermal.** Hot bright dip, near-boiling seal. *Defense:* face shield, heat gloves, steam awareness.
4. **Electrical & gas.** High-current DC; flammable hydrogen at the cathode. *Defense:* LOTO, dry hands, no ignition sources, ventilation.



REMEMBER

On most anodize lines the rule is “corrosives first.” On a *bright* line, add a rule above it: **toxic fume first**. The bright dip can hurt your lungs hours after a shift that felt fine. If you can name the hazard family at the tank in front of you — and you never, ever run the bright dip without ventilation — you already know most of what you need to protect yourself.



REMEMBER

You now have the whole foundation: what anodizing is (and how it's *not* plating), the part-is-the-anode/grow-oxide idea, the **brighten-then-clear-anodize** concept, the porous clear oxide, dimensional growth, the anodize family, **why bright demands high-purity aluminum**, how the line works, what to wear, and the safety mindset — **headlined by the bright-dip NO_x hazard**. Every station builds on this. Turn the page knowing the *why*, and the *how* will make a lot more sense.

PART TWO — STATION BY STATION
STATION 01 OF 12

INSPECTION & RACKING

“IN ANODIZING, THE PART IS THE WIRE. A LOOSE PART IS A DEAD PART — AND ON A BRIGHT JOB, THE WRONG METAL IS A DEAD JOB.”

PURPOSE & THE “WHY”

Inspect the incoming aluminum part, **confirm the alloy and purity** (mirror only happens on high-purity/bright alloys — Chapter 7) and the spec (bright/specular, gloss/DOI, thickness, clear or light tint, sealed), and **rack it** so it makes firm, current-carrying contact and hangs correctly. Because the part **is the anode**, every bit of current that grows your coating flows *through the rack contact into the part*. A loose or corroded contact means **no coating, a thin coating, or a burned contact spot** — and possibly a dropped part in a **hot acid tank**. Racking also decides *where* the un-anodized contact mark lands (it must be hidden on a mirror finish).



REMEMBER — CONFIRM THE METAL BEFORE ANYTHING ELSE

On a bright line, **inspection includes verifying the alloy/purity**. A casting or a high-Fe/Si alloy will never bright-anodize to a mirror no matter how perfectly you run the line. Catch the wrong metal here — not after you’ve processed a whole rack into scrap.

ITEM	SPEC / PRACTICE	WHAT TO WATCH
Incoming inspection	Per traveler / print	Confirm alloy & purity , bright/gloss spec, thickness, clear/tint, sealed
Surface condition	Polished/buffed or bright sheet	Scratches/gouges will stay — bright dip can’t fix them
Rack material	Titanium or aluminum	Ti racks don’t anodize away; Al racks anodize and need stripping
Contact pressure	Firm, spring-loaded / bolted	Loose = burning, no coat, dropped parts
Contact location	Hidden/non-critical area	The contact won’t anodize (visible bare mark on a mirror)
Masking	Plugs/tape/lacquer, clean edges	Threads, grounds, fits; clean edges so no witness line on a mirror
Orientation	Gas escape & drainage	Avoid trapped air (un-anodized voids); avoid bright-dip pooling



HEADS-UP

Racks and parts have **sharp edges and pinch points**, handled near **hot acid** tanks. A part dropped from a weak rack becomes a **hot-acid splash hazard**, not just scrap. Watch your hands and your footing.



TIP — TITANIUM RACKS ARE WORTH IT

Titanium racks don’t anodize (they passivate), so they keep good contact load after load and don’t need stripping. Aluminum racks anodize along with the part — insulating the contact over time — so they must be **stripped/re-pointed** regularly.

STEP-BY-STEP

1. Read the traveler — **confirm alloy/purity**, bright/gloss spec, thickness, clear/tint, masked areas. 2. Inspect for scratches/gouges — bright anodize **reveals and locks in** flaws, it doesn’t hide them. 3. Plan a **hidden** contact point that can carry full current. 4. Mask no-anodize areas with clean edges. 5. Rack firmly — good metal-to-metal contact, correct orientation. 6. Verify no parts touch (shadowing/marks). 7. Handle by the rack from here on — bare fingers leave prints that show on a mirror.

SIGN-OFF — STATION 01

Teach-Back: Why contact matters more in anodizing than plating (part = anode); why the contact point won’t anodize and why that’s worse on a mirror; **why you must verify alloy/purity for a bright job**; why titanium racks are preferred; two no-anodize areas that get masked and why.

“I verified the alloy/purity and bright spec, chose a firm hidden contact point, masked no-anodize features with clean edges, racked for gas escape and drainage with no parts touching, and handled parts by the rack.”

Trainee: _____ Trainer: _____ Date: _____

STATION 02 OF 12

CLEAN / DEGREASE

“GET IT CLEAN — BUT DON’T YOU DARE FROST IT.”

PURPOSE & THE “WHY”

Get the part **chemically clean** — no oil, grease, fingerprints, or shop soil — so the bright dip and oxide work evenly, using a **mild, non-etching cleaner** that **does not attack or matte the surface**. This is a key difference from a satin line: on a satin part the next step is an aggressive caustic etch, so cleaning is just cleaning. **On a bright line we must protect the polished surface**, so the cleaner is deliberately **gentle and non-etching** — it removes soil without dulling the metal. The free, foolproof check is the **water-break test**.



REMEMBER — NO FROSTING ALLOWED

A standard line can use a brisk alkaline (even mildly etching) clean because it’s about to matte-etch anyway. **A bright line cannot**. Use only the **non-etching cleaner** specified for bright work; a too-aggressive or too-hot caustic clean will **haze the surface and lower gloss** before the bright dip even gets a chance.



TIP — THE WATER-BREAK TEST (FREE, FAST, NEVER LIES)

Pull a cleaned part and watch the water. A truly clean surface sheets water in **one continuous, unbroken film** (PASS). A dirty surface makes water **bead up** (FAIL — re-clean). Costs nothing; takes two seconds; use it on every part. On a bright part it doubles as an early look at the surface.

FAIL - water beads up

```
+-----+
| o o o | X
| o o |
+-----+
```

Oil remains -> RE-CLEAN

PASS - water sheets evenly

```
+-----+
|#####| OK
|#####|
+-----+
```

Surface clean -> proceed

PARAMETER	RANGE	WHAT TO WATCH
Type	Mild, non-etching alkaline (or neutral) cleaner	Must not attack/matte aluminum
Temperature	~50–70 °C (120–160 °F)	Per cleaner TDS; not so hot it etches
Time	~3–10 min	Heavy oil longer
Concentration	Per TDS	Titrate regularly
Water-break test	Pass = continuous sheet	The reliable field check



HEADS-UP — HOT ALKALINE SOLUTION

Even a “mild” cleaner is hot and can be a skin/eye irritant; steam irritates airways. Splash goggles, gloves, apron. Skin → flush 15 min; eyes → eyewash 15 min and get help.

STEP-BY-STEP

1. Confirm temperature and concentration in range (and that it’s the **non-etching** product). 2. Immerse the racked load fully; start the timer; confirm agitation. 3. Withdraw slowly, draining back into the tank. 4. Run the **water-break test** — continuous sheet → proceed; beading → re-clean. 5. Move promptly toward the bright dip (don’t let a clean part sit and pick up airborne soil).

SIGN-OFF — STATION 02

Teach-Back: What the cleaner removes (oil/soil) and what it must **not** do (etch/frost the surface — and **why that’s critical on a bright line**); correctly perform/read a water-break test; the hazard and PPE.

“The part passed the water-break test, I used the non-etching cleaner in range, and I protected the polished surface.”

Trainee: _____ Trainer: _____ Date: _____

STATION 03 OF 12 • THE HEART

BRIGHT DIP — HOT PHOSPHORIC-NITRIC

WHERE ALUMINUM BECOMES A MIRROR — AND WHERE THIS MANUAL GETS DEADLY SERIOUS

Take a breath — and put your respect for this tank ahead of everything else. Up to now we’ve cleaned and racked. **Station 03 is where the brilliance is created** — and it is the single most hazardous tank in this book.

PURPOSE & THE “WHY”

Chemically level and brighten the surface to a specular mirror by immersing it in a hot, concentrated phosphoric-nitric acid bath that dissolves microscopic peaks faster than valleys (Chapter 3). The bright dip *replaces* the matte etch of a satin line — instead of frosting the surface, it **polishes** it. This is the step that makes “bright anodize” bright.

⚠

HEADS-UP — READ THIS BEFORE YOU READ ANYTHING ELSE ABOUT THIS TANK

The bright dip is **hot (~90–110 °C)** concentrated phosphoric + nitric acid. It causes **instant, severe chemical and thermal burns** on contact, and it evolves **toxic brown NOx (nitrogen-oxide) fumes**. NOx can injure your lungs hours after an exposure that felt minor (delayed pulmonary edema). **Never operate this tank without local exhaust / fume scrubbing confirmed running**. Full PPE: sealed goggles, face shield, heat- and acid-resistant gloves, apron, boots. On any fume exposure: fresh air, rest, **medical evaluation** — do not wait and see.

THE CHEMISTRY (KNOW WHAT’S IN THE TANK)

COMPONENT / PARAMETER	TYPICAL (VERIFY TDS — USUALLY PROPRIETARY)	WHAT IT DOES / WATCH
Phosphoric acid (H ₃ PO ₄)	The bulk of the bath (high %)	Viscous diffusion layer → leveling/smoothing
Nitric acid (HNO ₃)	A few percent	Oxidizer → brightening ; source of NOx fumes
(Sometimes) sulfuric acid	Small % (three-acid baths)	Modifies brightening; can lower nitric/NOx
Trace additives (Cu, surfactants)	Per proprietary product	Tune brightness, suppress fuming/pitting
Temperature	~90–110 °C (often ~95–105)	Hot — burns + drives fuming; too cold = poor bright
Immersion time	~30 s – 3 min	Too long = etch/over-attack, dimension loss, more fumes
Agitation	Gentle / per TDS	Even brightening; avoid splashing hot acid
Result	Specular, mirror surface + thin film	Smut is removed at desmut (Station 05)

★

REMEMBER — THE NUMBERS TO CARRY

Phosphoric-rich + a few % nitric, **~90–110 °C**, **~½–3 min**, **gentle agitation**, **ventilation ON**. And the safety number that matters most: **never zero ventilation**. Bright dip with no exhaust = NOx exposure.

⚙

TECHNICAL STUFF — HOW LEVELING ACTUALLY HAPPENS

Over the surface sits a **viscous, phosphate-rich diffusion film**. Microscopic *peaks* poke up through this film into fresher, more aggressive acid and dissolve faster than the *valleys*, which sit deeper in the sluggish film. Over seconds-to-minutes the peaks are shaved down and the surface **flattens to optical smoothness** — that’s *leveling*. The nitric simultaneously keeps the freshly exposed aluminum actively dissolving and bright (depolarizing) — that’s *brightening*. Leveling + brightening = mirror.

 **TECHNICAL STUFF — WHERE THE BROWN FUMES COME FROM**

When nitric acid attacks aluminum it is reduced, producing **nitrogen oxides (NO, which oxidizes to brown NO₂ in air)** — the visible brown/orange plume over the tank. More nitric, hotter bath, more active dissolution = more NOx. This is *inherent* to nitric bright dips. The controls are **engineering** (lateral/push-pull exhaust over the tank, fume scrubbers, bath additives that suppress fuming) — not PPE alone. Three-acid baths reduce, but don't eliminate, the fuming.

 **HEADS-UP — HOT-ACID SPLASH AND THE STEAM/MIST**

At ~90–110 °C the bath gives off **acid mist and steam** as well as NOx. Lower and lift parts **slowly and smoothly** — a fast plunge or yank throws hot acid. Keep your face out of the plume. Confirm the exhaust is pulling the fumes *away* from your breathing zone before you start.

STEP-BY-STEP

1. **Confirm ventilation/scrubber is running** and pulling fumes away — *no ventilation, no bright dip*. 2. Confirm bath temperature and chemistry are in range (per lab/TDS). 3. Confirm the part is clean (water-break pass) and the surface is sound (scratches won't be fixed). 4. Don full PPE for **hot acid + fumes**. 5. **Lower the rack slowly and smoothly** into the bath; start the timer. 6. Dwell the **specified short time** — just enough to reach target brilliance; over-dipping etches, loses dimension, and fumes more. 7. **Lift slowly and smoothly**, drain over the tank. 8. Move **directly to rinse** — don't let hot acid dry on the part. 9. Log time, temperature, and any observations.

COMMON DEFECTS — WHAT YOU'LL SEE

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Dull / no brightening	Bath too cold; time too short; depleted chemistry; wrong alloy	Check temp/time/chemistry; verify alloy/purity
Hazy / milky / frosty	High-Fe/Si alloy or casting ; over-etched; depleted bath	Confirm alloy; shorten dip; check bath
Orange-peel / wavy texture	Over-dipping; coarse grain; too aggressive	Shorten time; review incoming metal/temper
Pitting / gas streaks	Trapped gas; uneven agitation; impurities	Re-orient on rack; check agitation/bath
Heavy brown plume	Hot/active nitric (normal to a degree)	Verify exhaust/scrubber; report excessive fuming
Streaks / drip marks	Slow/uneven withdrawal; drying on part	Withdraw smoothly; rinse immediately

 **HEADS-UP — TEACH-BACK RED CARD: THE THREE WAYS THIS TANK HURTS YOU**

(1) **Hot-acid burns** (chemical + thermal) — full PPE, smooth handling. (2) **Toxic NOx fumes with delayed lung effects** — ventilation first, fresh air + medical evaluation on exposure. (3) **Splash/steam** — face shield, slow immersion. Memorize these before you run it.

 **FUN FACT**

Watch a part go in dull and come out in under a minute looking chrome-plated and it feels like a magic trick. It isn't — it's controlled, microscopic dissolution shaving down hills you can't see. The same brown breath that makes that magic is exactly the gas that demands the ventilation. Beauty and hazard, from the same bottle of nitric.

SIGN-OFF — STATION 03

*"I can independently: confirm the exhaust/scrubber is running before any work; verify bath temperature and chemistry; lower and lift parts smoothly with full hot-acid PPE; dwell only the specified time for target brilliance; recognize dull/hazy/orange-peel/pitting defects; and state the three hazards — hot-acid burns, toxic NOx fumes with **delayed** lung effects, and splash/steam — including the correct response to a fume exposure (fresh air + medical evaluation). I understand chemistry adjustments are made only by authorized personnel."*

Operator: _____ Date: _____ Trainer: _____ Date: _____

STATION 04 OF 12

RINSE (THE FIREWALL)

“A RINSE ISN’T A REST STOP. AFTER THE BRIGHT DIP, IT’S THE WALL THAT STOPS HOT ACID FROM FOLLOWING THE PART DOWNSTREAM.”

PURPOSE & THE “WHY”

Wash the **hot phosphoric-nitric film** off the part before it reaches the desmut, anodize, and seal chemistry. Bright-dip drag-out is aggressive, hot, and **NOx-bearing** — carry it forward and it contaminates baths, wastes chemistry, and can keep fuming. The rinse knocks drag-out down by dilution; cleanliness is tracked by **conductivity** (µS/cm) — *lower = cleaner*.



TIP — COUNTER-FLOW RINSING

A **counter-current** cascade (parts move forward, water flows backward, cleanest water touches the part last) saves enormous water on a line with this many rinses, and gives the cleanest handoff.

PARAMETER	RANGE	NOTES
Temperature	Ambient	No heating needed
Time	30–60 s with agitation	Longer for cavities/blind holes
Water	Municipal or DI	DI preferred near anodize/seal
Conductivity	Per shop spec (lower = cleaner)	Monitor at shift start
Flow	Continuous overflow / counter-current	Prevents stagnation



HEADS-UP

This rinse carries dragged-out **hot phosphoric-nitric acid** — acidic, possibly still off-gassing a little NOx right after the dip, and a **slip hazard**. Goggles, gloves, non-slip footwear; keep it under the ventilation zone. **Wet floors and slips are the #1 everyday injury at rinses.**

STEP-BY-STEP

- Confirm flow and conductivity.
- Transfer **promptly** from the bright dip — don’t let hot acid dry.
- Immerse fully with agitation 30–60 s (longer for cavities).
- Lift and drain over the rinse tank before moving on.

TOP MISTAKES

- Treating the rinse as a quick dunk.
- Ignoring conductivity.
- Letting parts drip-dry between bright dip and rinse.
- Not flushing cavities; skipping the drain step.

SIGN-OFF — STATION 04

Teach-Back: Why carrying hot bright-dip drag-out forward causes problems; define drag-out/drag-in; what conductivity tells you and which way is “clean.”

“Rinse conductivity was within limit, flow was running, I transferred promptly from the bright dip, and I rinsed the full time with PPE.”

Trainee: _____ Trainer: _____ Conductivity: _____ µS/cm

STATION 05 OF 12

DESMUT / DEOX

“THE BRIGHT DIP LEAVES A FAINT FILM. DESMUT WIPES IT OFF SO THE MIRROR — AND THE CLEAR OXIDE — COME OUT FLAWLESS.”

PURPOSE & THE “WHY”

Remove the thin **film/smut** the bright dip leaves behind, using a **nitric acid dip (deox/desmut)**, leaving a bright, uniform, chemically active surface for anodizing. The bright dip dissolves aluminum and can leave a light residue/oxide film and a slight surface smut; anodize over it and you risk **haze, streaks, or dull patches** — exactly what you can’t afford on a mirror finish. A short **nitric** dip clears it and “activates” the surface. (On higher-alloy parts a fluoride-bearing or proprietary deox may be specified — know which your tank is.)



REMEMBER

A faint film after the bright dip is **normal**. Desmut’s job is to remove it so the clear anodize grows water-clear over a clean mirror. A part that goes into the anodize tank still filmed will come out hazy — desmut is the **clarity gatekeeper**.

PARAMETER	RANGE / PRACTICE	NOTES
Chemistry	Nitric acid (or proprietary/fluoride deox for some alloys)	Match to the alloy/film
Concentration	Per product TDS	Titrate regularly
Temperature	Often ambient-warm	Per TDS
Time	~0.5–2 min	Just until film clears; no longer
Result	Bright, uniform, active surface	No haze/film remaining



HEADS-UP — NITRIC ACID (AND POSSIBLY FLUORIDE)

A nitric desmut is acidic and can **also evolve some NOx** if strong/warm — keep ventilation on. If your deox is **fluoride-bearing**, it’s especially hazardous (deep, delayed burns; systemically toxic) — read its SDS, wear full acid PPE, and know the **specific first-aid** (often **calcium gluconate** per your plan). **Add acid to water** for any make-up.

STEP-BY-STEP

1. Confirm which desmut chemistry the tank is (and its hazards/first-aid). 2. Check concentration/temperature in range; ventilation on. 3. Immerse the **short** specified time — just until the film clears. 4. Withdraw, drain, move to rinse. 5. Don’t over-dip — excessive desmut can etch/dull some alloys (the opposite of what we want).

SIGN-OFF — STATION 05

Teach-Back: What the bright-dip film is and why it must go before anodizing; that desmut chemistry depends on the alloy; the nitric (and possible NOx/fluoride) hazard and first-aid.

“I cleared the bright-dip film with the correct desmut, did not over-dip, kept ventilation on, and understand the specific hazard/first-aid for this tank.”

Trainee: _____ Trainer: _____ Chemistry: _____

STATION 06 OF 12

RINSE (PRE-ANODIZE)

“THE LAST CLEAN HANDOFF BEFORE THE MAIN EVENT.”

PURPOSE & THE “WHY”

Remove desmut acid before the part enters the anodize tank. Carrying desmut drag-out into the anodize bath introduces unwanted ions (and, with a fluoride desmut, **fluoride contamination** that can roughen/pit the anodize and ruin a mirror). A clean rinse protects the anodize chemistry and the finish.

PARAMETER	RANGE	NOTES
Temperature	Ambient	—
Time	30–60 s, agitation	Longer for cavities
Water	DI preferred	Cleaner handoff to anodize
Conductivity	Per shop spec (lower = cleaner)	—
Flow	Counter-current overflow	—



HEADS-UP

Same rinse rules: drag-in acid, wet floors, goggles/gloves. If the desmut was **fluoride-bearing**, this rinse water carries fluoride — handle and dispose accordingly.



TIP

Move from this rinse into the anodize tank **promptly and consistently**. A long, variable dwell in air before anodizing causes part-to-part variation — and on a bright job, consistency *is* quality.

STEP-BY-STEP

- Confirm flow and conductivity.
- Rinse the full time with agitation.
- Drain over the tank.
- Move straight to anodize.

TOP MISTAKES

- Quick dunk instead of a real rinse.
- Inconsistent dwell before anodize.
- Ignoring conductivity.
- Forgetting fluoride drag-in handling.

SIGN-OFF — STATION 06

Teach-Back: Why a clean handoff to the anodize tank matters; what fluoride drag-in could do to a mirror finish.

“The part was fully rinsed, conductivity acceptable, and I moved promptly and consistently to the anodize tank.”

Trainee: _____ Trainer: _____ Date: _____

STATION 07 OF 12 • THE MAIN EVENT

CLEAR ANODIZE — THE MAIN TANK

A THIN, WATER-CLEAR OXIDE GROWN FROM THE PART’S OWN SURFACE — SEALING THE MIRROR UNDER GLASS

Up to now every station has prepared a perfect mirror. **Station 07 grows the clear, transparent oxide that protects it forever** — kept thin and clear on purpose so the brilliance shows right through.

The big idea in one sentence: **we run the bright-dipped aluminum part as the *anode* in cold, dilute sulfuric acid and use DC to grow a THIN, TRANSPARENT aluminum-oxide layer out of the part’s own surface — clear enough to see the mirror through.** The cast:

- **Anode** — *your aluminum part (positive)*. Where the oxide grows.
- **Cathode** — **lead or aluminum (6063)** plates (negative). They carry current and bubble off hydrogen; they do **not** feed metal into the part.
- **Electrolyte** — **dilute sulfuric acid (~165–200 g/L)**, held **cold (~20–21 °C / 68–72 °F)** with **air agitation** and **refrigerated cooling**.

★

REMEMBER — SAME TANK AS TYPE II, RUN FOR CLARITY

This is fundamentally a **Type II sulfuric anodize** — same chemistry, same polarity (**part = anode**), same controls. The bright twist is **how** we run it: **thin** (short time / controlled thickness) and scrupulously **clean and cold**, so the oxide stays **transparent** and the mirror shows through. Heat, contamination, or excess thickness all add **haze** — the enemy of a bright finish.

⚠

HEADS-UP — SULFURIC ACID + DC + HYDROGEN + COOLING, ALL AT ONCE

Strong **sulfuric acid** (burns, splash, mist), **high-current DC** (shock/arc at contacts), **flammable hydrogen** off the cathodes, and a bath that **self-heats** (cooling must work or the coating burns and hazes). Engineering controls — **exhaust, cooling, agitation, splash control** — must be working before you run it. Full acid PPE. No ignition sources.

• WHY THIN & CLEAR (AND WHAT MAKES IT PARTICULAR)

GOAL	WHAT IT MEANS ON THE FLOOR
Transparent oxide	On high-purity Al, the oxide is water-clear — mirror shows through
Thin coating	Less light scatter → higher gloss/DOI; runs short
Cold, clean bath	Warm/dirty bath = haze, burn, dullness — fatal on a mirror
Sealable	Pores still close in seal to lock in brilliance/corrosion resistance

...but the same chemistry demands care: it must be **kept cold** (warm = soft/powdery/**hazy**), it needs **steady current** (thickness/clarity depend on it), it’s sensitive to **chloride** (pitting — ruins a mirror), and it grows the coating **half into the part** (dimensional growth).

• THE BATH MAKE-UP — KNOW THE RANGES

**HEADS-UP**

These are **typical** clear-anodize ranges for training context. **The controlling document is your customer/process spec — ASTM B580, MIL-PRF-8625 Type II, and your shop’s lab sheet.** Verify before you run; never anodize to a number from a training manual.

COMPONENT / PARAMETER	TYPICAL RANGE	WHAT IT DOES / WATCH
Sulfuric acid (H ₂ SO ₄)	~165–200 g/L (~15–20% v/v)	The electrolyte; grows + gently dissolves the oxide
Temperature	~20–21 °C (68–72 °F)	The critical control — too warm = soft/hazy/burned
Voltage (DC)	~12–18 V	Driven to hold target current density
Current density	~12–18 ASF (~1.2–2.0 A/dm ²)	Sets growth rate & structure
Coating thickness	~3–8 µm (kept thin)	Thinner preserves gloss; per spec
Time	Short (~10–25 min typical)	Long = thicker = duller
Cathodes	Lead or aluminum 6063	Carry current; bubble H ₂ ; don’t feed metal
Agitation	Air agitation	Even temperature & fresh acid at the surface
Cooling	Refrigeration / heat exchanger	Bath self-heats — <i>must</i> be cooled
Dissolved aluminum (Al ³⁺)	Per spec (kept moderate)	Too much dulls/hazes & slows the bath
Chloride contamination	Very low (≤ ~50 ppm)	Causes pitting — fatal to a mirror

**REMEMBER — THE NUMBERS TO CARRY**

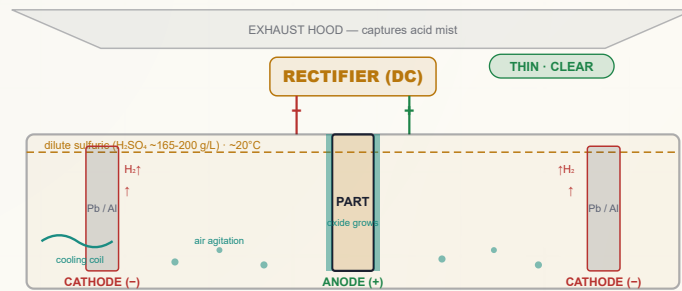
Sulfuric ~165–200 g/L, ~20–21 °C, ~12–18 V, ~12–18 ASF, THIN ~3–8 µm, short time, lead/6063 cathodes, air agitation + cooling. And the bright-line emphasis: **thin, cold, clean = clear.**

**TECHNICAL STUFF — THICKNESS MATH (THE “720 RULE”)**

Thickness is set by **current density × time**. A shop rule: *minutes* ≈ (720 × *thickness in mils*) ÷ ASF. For thin bright work you’re targeting only ~0.1–0.3 mil, so runs are **short**. Verify with an eddy-current gauge — but resist the urge to “give it a little extra,” because **extra thickness costs gloss.**

**TECHNICAL STUFF — TEMPERATURE IS KING, AND DISSOLVED AL / CHLORIDE**

Warm bath → acid dissolves the growing oxide too fast → soft, powdery, **hazy**/“**burned**” coating — instantly visible scrap on a mirror. Hold ~20–21 °C tightly. A little **dissolved aluminum** helps; **too much** dulls/hazes and slows growth. **Chloride** (tap water, sweat, drag-in) causes **pitting** — and a single pit ruins a mirror. Use **DI water** and keep contamination out.



Key params: H₂SO₄ 165-200 g/L • 20-21C • 12-18 V • 12-18 ASF • THIN 3-8 um • clear. Note the **reversed polarity vs. a plating cell**: the part is the ANODE (+).

• STANDARD OPERATING PROCEDURE

1. **Confirm the part is clean, bright-dipped, desmutted, rinsed, and racked tight** — straight from the rinse, no long air dwell.
2. **Verify engineering controls FIRST**: cooling at temp (~20–21 °C), air agitation on, exhaust on, splash control, full acid PPE.
3. **Check bath readiness**: sulfuric in range, temperature in range, dissolved-Al and chloride within limits, cathodes sound.
4. **Confirm rectifier polarity: PART = ANODE (POSITIVE)** — *opposite of plating; verify every time.*
5. **Set current** for the part's surface area to hit ~12–18 ASF.
6. **Calculate the short time** for the thin target thickness; set timer/amp-hour counter.
7. **Energize and ramp**; watch voltage rise to hold current as the oxide grows (normal).
8. **Anodize the calculated (short) time**, monitoring temperature, current/voltage, agitation, cooling.
9. **At target, stop current, withdraw, drain over the tank**, move to the **cool DI** post-anodize rinse — pores now open.
10. **Log everything.**



HEADS-UP — VERIFY POLARITY EVERY SINGLE TIME

Everyone trained on plating expects the part to be the **cathode**. **The part is the ANODE (+)**. Reversed polarity gives no coating and can damage parts. Check before you energize.

COMMON DEFECTS — WHAT YOU’LL SEE AND WHAT IT MEANS

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Haze / loss of gloss	Bath warm; coating too thick; high dissolved Al; impure alloy	Cool bath; reduce thickness; lab-check Al; verify alloy
Powdery / soft / “burned”	Overheated bath; CD too high	Restore cooling/agitation; lower CD
Thin / no coating	Bad contact; wrong polarity ; low current/acid	Re-rack; verify part = anode ; check current/acid
Pitting	Chloride contamination	Lab-check chloride; use DI; treat bath
Cloudy/gray cast	High-Si/Cu or low-purity alloy	Confirm alloy/purity (Chapter 7)
Voltage spikes / won’t hold current	Poor contact; failing cathodes	Check contacts/cathodes; lab-check bath

INTEGRATED SAFETY — ACID, DC, HYDROGEN, COOLING



HEADS-UP — WATCH COOLING THE WHOLE RUN

The bath self-heats. If cooling fails mid-run, the coating **hazes/burns** and the bath degrades. Don’t set-and-forget — monitor temperature, current/voltage, and agitation the entire cycle.



HEADS-UP — STRONG SULFURIC + HIGH-CURRENT DC + HYDROGEN

Sulfuric burns (splash/mist) — full acid PPE, 15-minute flush, **acid to water**. LOTO for all electrical work, dry hands only. Hydrogen off the cathodes is **flammable** — ventilation on; no flames, sparks, or grinding nearby.

TEACH-BACK — CAN YOU ANSWER?

- Which electrode is your part, and which charge? (*Anode, +.*)
- Where does the coating come from? (*The part’s own Al + O₂.*)
- The headline numbers (acid, temp, V, ASF, thickness, time)?
- Why keep it thin, cold, and clean? (*Clarity = brilliance.*)
- What does chloride do? (*Pitting — ruins a mirror.*)
- The acid, electrical, and hydrogen hazards?

TOP MISTAKES TO AVOID

- Assuming the part is the cathode (it’s the **anode**).
- “A little extra thickness” (costs gloss).
- Letting the bath warm or cooling fail.
- Loose rack contact; ignoring chloride.
- Tap-water drag-in (pitting/haze).
- Set-and-forget instead of monitoring.

SIGN-OFF — STATION 07

“I can independently: verify cooling, agitation, exhaust, and PPE before running; verify bath readiness (acid, temp, dissolved Al, chloride); confirm the part is racked as the ANODE (positive); run a THIN, CLEAR coating by controlling current × time; recognize haze/burn/pitting/thin-coat defects; and state the acid, electrical, and hydrogen hazards. I understand chemistry adjustments are made only by authorized personnel.”

Operator: _____ Date: _____ Trainer: _____ Date: _____

STATION 08 OF 12

RINSE (POST-ANODIZE)

“GENTLE NOW — THE PORES ARE WIDE OPEN, AND THE MIRROR IS ONE WATER SPOT FROM A COMPLAINT.”

PURPOSE & THE “WHY”

Rinse the sulfuric acid out of the freshly anodized part **without contaminating the open pores or pre-sealing them**, before seal (or an optional light tint). The fresh oxide is **porous and absorbent** — it soaks up whatever it touches. Leftover acid interferes with sealing/tinting; impurities cause spots and haze. Use **clean DI**, kept **cool** — warm water here can begin to seal the pores early and (if a tint is wanted) block uptake, or cause uneven sealing.



REMEMBER — DON'T PRE-SEAL OR SPOT BY ACCIDENT

After anodizing, use **cool/ambient DI** rinse. Hot or hard water here can start closing pores unevenly or leave mineral spots — and on a mirror, spots and uneven seal show instantly. Save the heat for the actual seal step.

PARAMETER	RANGE	NOTES
Water	DI strongly preferred	Pores absorb impurities; spots show on a mirror
Temperature	Cool / ambient	Avoid premature/uneven sealing
Time	1–3 min, gentle agitation	Flush acid from pores
Conductivity	Low (per spec)	Confirms acid removed



HEADS-UP

Carries dragged-out **sulfuric acid** — mildly acidic, slip hazard, goggles/gloves. Don't send acid-laden rinse untreated to drain (Part Three / waste).

STEP-BY-STEP

- Confirm DI flow, cool temperature.
- Rinse gently, full time.
- Drain.
- Move to seal (or, if a light tint is specified, to dye first).

TOP MISTAKES

- Using hot water before sealing (pre-seals pores).
- Using hard tap water (spots/impurities in pores).
- Short rinse leaving acid behind.

SIGN-OFF — STATION 08

Teach-Back: Why the pores are absorbent now; why cool DI before seal/tint; what hot or hard water would do to a mirror finish.

“I rinsed with cool DI to flush acid from the open pores without pre-sealing or spotting, and moved promptly to seal (or tint).”

Trainee: _____ Trainer: _____ Date: _____

— STATION 09 OF 12 • USUALLY SKIPPED

DYE (LIGHT TINT ONLY)

“ON A BRIGHT JOB, COLOR IS THE EXCEPTION — AND WHEN USED, IT’S A WHISPER, NOT A SHOUT.”

PURPOSE & THE “WHY”

Only if a **light, transparent tint** is specified, draw a delicate color into the open pores (e.g., bright gold/bronze decorative trim). **Most bright work is left clear** because the whole point is brilliance, and heavy color muddies the shine. A thin clear coat has few pores, so it takes only **light, transparent shades** — dark/opaque colors defeat the purpose and would require a thicker (duller) coat. If tinted, the part **must then be sealed** to lock the color in.



REMEMBER — CLEAR IS THE DEFAULT; TINT THEN SEAL

For most bright parts: **skip dye, go straight to seal**. When a tint is specified: **anodize → cool DI rinse → light tint → rinse → seal**, never reversed. Once sealed, you can’t tint; once tinted, you must seal or it bleeds/fades.

PARAMETER	RANGE / PRACTICE	NOTES
Use	Usually none (clear); else light transparent dye	Brilliance is the goal
Dye type	Light-fast organic / metallic salt for trim	Choose UV-stable for exterior
Temperature	~50–60 °C typical	Per dye TDS
pH	Per dye (often slightly acidic)	Critical for shade/consistency
Time	Short – to a light shade	Don’t over-dye (muddies shine)
Water	DI	Tap impurities ruin color & gloss



HEADS-UP — DYE HANDLING

Some dyes are skin/eye irritants or sensitizers; baths may be warm and mildly acidic. Gloves, goggles, read each dye’s SDS (some carry their own disposal requirements).



TIP — CONSISTENCY IS EVERYTHING IN COLOR

Light tints show variation badly. Keep coating thickness, dwell, temperature, pH, and the anodize-to-tint interval consistent part-to-part, and dye promptly after the post-anodize rinse.



TECHNICAL STUFF — WHY A THIN CLEAR COAT TAKES ONLY LIGHT SHADES

Color depth depends on how much dye the pores can hold — and a thin bright anodize has a shallow porous layer. There simply isn’t enough pore volume for deep, opaque color without going thicker (which dulls the mirror). So bright tinting is inherently **delicate and transparent** — a light wash of color over a shine, not a solid hue.

SIGN-OFF — STATION 09

Teach-Back: Why most bright work is clear; correct order if tinting; why only light/transparent shades suit a thin bright coat; dye hazards.

“I either left the part clear or applied only a light transparent tint before sealing, into open pores, with consistent parameters in DI.”

Trainee: _____ Trainer: _____ Tint/none: _____

STATION 10 OF 12

SEAL (CLEAR — PRESERVE THE BRILLIANCE)

“THE FINAL LOCK — CLOSE THE PORES TO KEEP THE MIRROR BRIGHT AND THE CORROSION OUT, WITHOUT A SINGLE HAZY FILM.”

PURPOSE & THE “WHY”

Close the open pores of the clear oxide so the coating reaches full corrosion resistance and the brilliance is locked in permanently — and do it without any surface film/bloom that would dull the mirror. An unsealed anodize is porous: it absorbs stains, water-spots, and corrodes through the pores. Sealing swells/plugs the pore mouths shut. The classic method is hot DI water near boiling (forms boehmite); there are also mid-temperature nickel-acetate and cold nickel-fluoride seals. On bright work, choosing a seal that stays clear and bloom-free is critical.



REMEMBER — ON A MIRROR, BLOOM IS THE ENEMY

A “sealing bloom” (chalky/smudgy film from hard water, wrong pH, or over-concentration) is a minor blemish on a satin part but ruins a bright part. Use DI water, correct pH/temperature, and clean chemistry — and inspect for bloom every time. Sealing is where a brilliant job is finished — or quietly dulled.

SEAL TYPE	TEMPERATURE	TRADEOFFS (BRIGHT-LINE VIEW)
Hot DI water	~96–100 °C	Simple, metal-free, clear; high energy, slow, bloom-prone if water isn’t DI
Nickel-acetate (mid-temp)	~80–90 °C	Efficient, excellent corrosion/retention, generally clear; contains nickel
Cold nickel-fluoride	~25–30 °C	Fast/low-energy; needs tight control or it blooms; contains nickel + fluoride



HEADS-UP — NEAR-BOILING TANK = SCALD HAZARD

Hot-water or hot nickel seals run ~80–100 °C. Splashes and steam cause serious scald burns. Face shield, heat-resistant gloves, long sleeves; lower parts gently — a fast plunge throws hot water.



HEADS-UP — NICKEL SEALS ARE A HEALTH & WASTE CONCERN

Nickel is a skin sensitizer; the fluoride in cold seals is especially hazardous (deep, delayed burns; systemic toxicity). Gloves and goggles, read the SDS, know the fluoride first-aid (calcium gluconate per your plan), and route nickel/fluoride waste to proper treatment — never to drain.

STEP-BY-STEP

1. Confirm seal type, temperature, pH, and DI water in range. 2. Confirm the part came from a clean cool rinse (no acid/contamination). 3. Immerse gently (mind the scald), full time for the thin coating (hot-water seal ~2–3 min per µm — thin coats seal quickly). 4. Withdraw, rinse if specified, and inspect for bloom under good light. 5. Move to final rinse/dry.

SIGN-OFF — STATION 10

“I sealed at the correct type/temperature/pH in DI water, inspected for bloom (critical on a mirror), and handled the scald and (if applicable) nickel/fluoride hazards correctly.”

Trainee: _____ Trainer: _____ Seal type/temp: _____

STATION 11 OF 12

RINSE / DRY

“FINISH CLEAN, DRY SPOT-FREE, DON’T FINGERPRINT YOUR OWN MIRROR.”

PURPOSE & THE “WHY”

Give the sealed part a final **DI** rinse and a **gentle, spot-free dry**. A bright mirror finish shows **every water spot and fingerprint**, so the final rinse must be clean and the dry must be gentle and even. Aggressive heat or hard-water spotting at the very end blemishes an otherwise perfect mirror.

PARAMETER	RANGE	NOTES
Final rinse water	DI	Spot-free is mandatory on a mirror
Rinse temperature	Warm	Aids fast, even drying
Dry method	Air / warm forced air; gentle oven	Avoid excessive heat; even drying
Handling	By rack / clean gloves	No bare-hand prints on finished mirror parts

**HEADS-UP**
Parts off a hot seal are **hot** — burn risk. Let them cool or use gloves. Dry ovens are a heat hazard; mind hot racks.

**TIP**
A **DI final rinse** is the cheapest insurance against water spots on a finished mirror. Tap-water spots on a bright clear anodize are glaringly obvious and are pure rework.

STEP-BY-STEP

- Final DI rinse.
- Drain.
- Gentle, even dry.
- Handle by rack/clean gloves to inspection.

TOP MISTAKES

- Tap-water final rinse (spots).
- Over-aggressive drying.
- Bare-hand handling of a finished mirror.**
- Uneven drying leaving streaks.

SIGN-OFF — STATION 11

Teach-Back: Why a clean DI final rinse + gentle even dry protects a mirror finish; why finished bright parts are handled only by rack/gloves.

“I gave a clean DI final rinse, dried gently and spot-free, and handled finished parts cleanly.”

Trainee: _____ Trainer: _____ Date: _____

STATION 12 OF 12

FINAL INSPECTION

“PROVE IT’S BRILLIANT — GLOSS, DISTINCTNESS-OF-IMAGE, THICKNESS, SEAL, AND FINISH.”

PURPOSE & THE “WHY”

Verify the finished part meets spec: **specular gloss and distinctness-of-image (the brilliance)**, **coating thickness**, **seal quality**, and **overall appearance**. On a bright job, **brilliance is the deliverable**, so it’s measured, not just eyeballed: **gloss** with a gloss meter and **DOI** for image sharpness. **Thickness** confirms the (thin) oxide grew; **seal quality** confirms the pores closed; and a visual check catches haze, bloom, pits, scratches, contact marks, and handling damage.

CHECK	METHOD	PASS CRITERIA
Specular gloss	Gloss meter (ASTM D523, 20°/60°)	Meets gloss-unit spec
Distinctness-of-image (DOI)	ASTM E430 / visual reflected-image check	Sharp, undistorted reflected image
Coating thickness	Eddy-current gauge (ASTM B244)	Within (thin) spec
Seal quality	Dye-stain (ASTM B136); admittance (ASTM B457)	Low pickup/admittance = well sealed
Appearance	Visual under good light	No haze, bloom, pits, scratches, bad marks
Dimensions	Gauge where critical	Within tolerance after growth (+ bright-dip loss)



REMEMBER — THE CHECKS THAT MATTER MOST ON A BRIGHT JOB

Gloss/DOI = “is it brilliant?”, eddy-current = “right (thin) thickness?”, seal test = “pores closed?”. A bright part can be the right thickness and well sealed and **still fail** if it isn’t bright enough — which is exactly why gloss/DOI are measured.



TIP — THE REFLECTED-IMAGE CHECK IN PLAIN TERMS

Hold the part so it reflects a sharp object (a light, a ruler, text). **High DOI** = the reflection is crisp and undistorted (true mirror). **Low DOI** = the reflection is blurry/cloudy (haze, orange-peel, or under-brightening). It’s the quick visual cousin of the gloss-meter and DOI instruments.

STEP-BY-STEP

- Measure gloss/DOI on significant surfaces.
- Verify thickness (eddy-current).
- Run the seal test.
- Visual for haze/bloom/pits/scratches/marks.
- Gauge critical dimensions; accept/reject/rework.

TOP MISTAKES

- Eyeballing brilliance instead of measuring.
- Checking only easy surfaces.
- Missing a faint bloom under poor light.
- Bare-hand handling of finished mirrors.
- Ignoring bright-dip metal loss on fits.

SIGN-OFF — STATION 12

Teach-Back: Which check answers “brilliant?” vs. “enough oxide?” vs. “pores closed?”; what a poor reflected-image (DOI) result tells you.

“I measured gloss/DOI, verified thickness, ran the seal test, inspected for haze/bloom/pits/scratches, and confirmed critical dimensions.”

Trainee: _____ Trainer: _____ Gloss/DOI: _____ · Thickness: _____ μm · Seal: P/F

PART THREE — ACROSS THE LINE
THE CROSS-CUTTING TOPICS

ACROSS THE LINE

THE IDEAS THAT DON'T LIVE IN ONE TANK — THEY TIE THE WHOLE LINE TOGETHER

• BRIGHTNESS & GLOSS MEASUREMENT (DEEP DIVE)

Brilliance is the product, so it's measured. **Specular gloss** (ASTM D523) uses a gloss meter at fixed angles (commonly **20°** for very high gloss and **60°** for general) reporting **gloss units** — higher = more mirror-like. **Distinctness-of-image (DOI)** (ASTM E430) measures how **sharp** a reflected image is — the truest measure of “mirror” quality, because a surface can be glossy yet slightly cloudy. **Total/specular reflectance** matters for reflector work. Track gloss/DOI against an approved **bright standard**, and remember the chain: **substrate purity** → **mechanical prep** → **bright dip** → **thin clear anodize** → **bloom-free seal** — every link can cost gloss.



REMEMBER

Two numbers define a bright job: **gloss units (ASTM D523)** and **DOI (ASTM E430)**. If you can't quantify the shine, you can't prove the job.

• THE BRIGHT DIP & ITS NOX HAZARD (DEEP DIVE — THE HEADLINE)

The hot phosphoric-nitric bright dip is the heart of the finish and the standout hazard. It **levels by preferential peak dissolution** and **brightens via nitric oxidation** — and that nitric inevitably evolves **brown NOx fumes** that are **toxic with delayed lung effects**. Controls, in order: **engineering first** — local/lateral or push-pull **exhaust** over the tank, **fume scrubbers**, bath additives that **suppress fuming**, lids, and not over-heating/over-dipping; **then** PPE and respiratory protection per program; **then** procedures (smooth immersion, short dwell, prompt rinse). **Never run it without ventilation**. On any suspected exposure: fresh air, rest, **medical evaluation** — NOx pulmonary edema can appear hours later.



HEADS-UP

The bright dip's brilliance and its toxic fume both come from the **nitric acid**. You cannot have one without managing the other. Engineering controls — exhaust and scrubbing — are non-negotiable, not optional add-ons.

• ALUMINUM PURITY & ALLOY EFFECTS (DEEP DIVE)

The single biggest variable. **Super-purity (99.99%)** and **bright alloys (5657/5252/5357/6463)** give true mirrors; **6063** is good; **2xxx (copper)** dulls/yellows; **cast/high-Si alloys cannot go bright**. Iron/silicon/copper/manganese form **light-scattering constituent particles** that haze the finish. Rack **like alloy with like**, and **anodize a sample of the actual lot** before promising brilliance. (Cross-ref Chapter 7.)

• AVOIDING THE MATTE ETCH (DEEP DIVE)

A standard satin line **wants** the matte caustic etch — it frosts the surface to the familiar satin look. **A bright line must avoid it entirely.** Cleaning is **non-etching**; the bright dip *polishes* rather than frosts. Any accidental caustic attack (too-hot cleaner, caustic drag-in, a stray etch dip) **frosts the mirror** and is unrecoverable short of re-polishing. Keep caustic chemistry off this line, or strictly controlled and downstream of nothing bright.

• COATING THICKNESS & DIMENSIONAL GROWTH (DEEP DIVE)

Thickness = current density × time — but here we run it **thin (~3–8 µm)** because thicker oxide **scatters light and lowers gloss**. Growth is **~half-in/half-out**, so a 5 µm coat moves each surface out ~2.5 µm. **Also account for metal removed by the bright dip** (it dissolves a little aluminum to level). Verify thickness by **eddy-current** — and resist “extra for safety,” which costs brilliance.

• RACKING & ELECTRICAL CONTACT (DEEP DIVE)

The part is the anode, so **the rack is the wire**. Firm current-carrying contact; **hidden** contact placement (a bare mark is glaring on a mirror); orientation for **gas escape and bright-dip drainage**; no parts touching; titanium racks preferred. Poor contact = thin coat, burned spots, dropped parts in **hot acid**. (*Cross-ref Station 01.*)

• SEAL CHOICE FOR CLARITY (DEEP DIVE)

Close the pores **without bloom**. **Hot DI water** (metal-free, clear, but bloom-prone with hard water), **nickel-acetate** (efficient, generally clear, contains nickel), **cold nickel-fluoride** (fast, but bloom-prone if uncontrolled, nickel+fluoride hazard). The universal enemies — **hard water, wrong pH, contamination** — cause **bloom**, which on a mirror is scrap. Confirm with **dye-stain (ASTM B136)** or **admittance (ASTM B457)**.

• RINSING & WATER / DI MANAGEMENT

Rinses are firewalls between chemistries and protectors of the finish. **Counter-current cascade** saves water; **conductivity** is the cleanliness gauge (lower = cleaner). Use **DI** for post-anodize rinse, any tint, seal, and final rinse — tap-water **chloride** and hardness cause **pitting, spots, and bloom**, all of which destroy a mirror.



REMEMBER

Across the whole line, four ideas rule a bright job: **high-purity metal**, a **well-ventilated bright dip**, a **thin/cold/clean clear anodize**, and a **bloom-free seal** — with **DI water and clean handling** protecting the mirror at every rinse in between.

HANDLE IT RIGHT

WASTE & ENVIRONMENTAL

AN ATTRACTIVE FINISH, A SERIOUS WASTE PROFILE — NEVER TO A FLOOR DRAIN OR OPEN AIR UNTREATED

A bright anodize line generates several regulated streams — **never to a floor drain or open air untreated**:

- **Spent phosphoric-nitric bright dip & its rinses** — strong acid; high **phosphate** and **nitrate (nitrogen)** load; dissolved **aluminum**. Neutralize/treat per permit; phosphate and nitrogen discharge limits are common and strict; spent baths are often **hailed** as hazardous waste.
- **NOx air emissions (the bright dip)** — brown nitrogen-oxide fumes require **local exhaust and fume scrubbing** (and possibly NOx-specific controls) per air permit. This is the standout *air* concern of the line.
- **Nitric desmut waste** — acidic, nitrate-bearing (and **fluoride** if a fluoride deox is used) — treat accordingly.
- **Spent sulfuric acid & acid rinses (anodize)** — neutralize/treat; manage **dissolved aluminum** (precipitated as aluminum hydroxide sludge).
- **Dye/tint waste (if used)** — colorants/regulated metals — treat/dispose per SDS and permit.
- **Nickel-seal waste** (nickel-acetate / nickel-fluoride) — **nickel is regulated**; fluoride needs treatment; segregate — never to sewer.



HEADS-UP

Acidic streams must be **neutralized under control** (never casually mixed), **phosphate and nitrogen** loads require dedicated treatment to meet discharge limits, and **NOx air emissions require working scrubbing/exhaust** per permit. Nickel/fluoride streams need dedicated treatment. Know your shop's segregation scheme — the wrong drum, drain, or an idle scrubber can be a reportable violation.



REMEMBER

The bright line's environmental headline is the **bright dip**: phosphate + nitrate in the water and **NOx in the air**. Both demand dedicated controls. Beauty out front, real responsibility out back.



FUN FACT

The phosphate dragged out of a bright-dip line is the same nutrient that feeds algae blooms — which is exactly why phosphate discharge limits exist. Good rinsing and segregation aren't just housekeeping; they keep that nutrient out of the watershed.

— HOW WE PROVE IT

QUALITY CHECKS & TEST METHODS

THE CHECKS THAT CONFIRM A BRIGHT PART IS RIGHT — ABOVE ALL, THAT IT'S BRILLIANT

CHECK	METHOD (TYPICAL STD)	ANSWERS
Specular gloss	Gloss meter (ASTM D523, 20°/60°)	"Is it brilliant?"
Distinctness-of-image	ASTM E430 / reflected-image	"Is it a true mirror?"
Coating thickness	Eddy-current (ASTM B244); micro-section	"Enough (thin) oxide?"
Seal quality	Dye-stain (ASTM B136); admittance (ASTM B457)	"Pores closed?"
Color (if tinted)	Visual / spectrophotometer vs. standard	"Right light shade?"
Corrosion	Salt spray (ASTM B117) per spec	"Will it survive service?"
Appearance/defects	Visual under controlled light	Haze, bloom, pits, scratches, marks

Standards note: overall anodic coatings per **ASTM B580** (architectural/decorative classes); clear-anodize portion often referenced to **MIL-PRF-8625 Type II**; bright-dip chemistry is usually proprietary — follow supplier TDS.

**TIP**

On a bright job the **gloss/DOI numbers are the deliverable** — not just a thickness reading. A part can be the right thickness and well sealed and still be rejected for not being bright enough. Measure the shine, every time, against the approved standard.

**REMEMBER**

The two bright questions in test form: **is it brilliant?** (gloss/DOI) and **is it sound?** (thickness, seal, no defects). Appearance under good light, plus the gloss meter, settles both — run the checks your customer spec actually requires.

**TECHNICAL STUFF — GLOSS VS. DOI, WHY BOTH**

A gloss meter measures how much light reflects *specularly* at a set angle; a surface can score high gloss yet still look slightly cloudy. DOI measures how *sharply* an image survives the reflection — it catches the subtle haze and orange-peel that gloss alone misses. For true mirror work (reflectors, premium trim), you want **both** high.

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

DEFECT / SYMPTOM	LIKELY CAUSE	FIX
Dull / not bright	Bright dip too cold/short/depleted; wrong (impure/cast) alloy ; over-thick anodize	Check bright-dip temp/time/chemistry; verify alloy/purity ; run anodize thinner
Hazy / milky / cloudy	Impure alloy; warm anodize bath; high dissolved Al; bloom from seal	Confirm alloy; cool/clean anodize; DI seal, correct pH
Orange-peel / wavy	Over-dipping in bright dip; coarse grain/temper in metal	Shorten bright dip; review incoming metal/temper
Frosted / matte patches	Accidental caustic attack (hot cleaner/etch drag-in); over-etch	Eliminate caustic exposure; use non-etching clean
Burning (powdery, rough, hazy)	Anodize bath warm; CD too high; poor agitation	Restore cooling/agitation; lower CD
Pitting	Chloride contamination ; bright-dip gas streaks	Lab-check chloride; use DI; fix agitation/orientation
Sealing bloom (chalky film)	Hard water; wrong pH; over-concentrated seal	DI water; correct pH/concentration; light post-dip
Water spots / fingerprints	Tap-water final rinse; bare-hand handling	DI final rinse; handle by rack/gloves
Contact burns / bare mark visible	Loose contact; mark placed on show surface	Tighten/relocate contact to hidden area
Scratches in finish	Pre-existing — bright dip can't repair	Improve upstream polish/handling
Gray cast on cast parts	High-silicon alloy (inherent)	Set expectations; bright not possible on castings

 **TIP**

Most bright-anodize defects trace to one of four roots: **substrate** (wrong/impure alloy, scratches → dull/hazy/frosty), **bright dip** (temp/time → dull or orange-peel), **anodize** (warm/thick/dirty → haze/burn/pits), or **seal/handling** (bloom, spots, prints). Name the root, not just the symptom.

 **REMEMBER**

Two of these are **safety-relevant** as well as quality: **wrong polarity** (verify part = anode) and **running the bright dip without ventilation** (NOx exposure). When in doubt about a bath change, ask your supervisor before you act.

— EVERY TERM, PLAIN AND SIMPLE

GLOSSARY

PLAIN-LANGUAGE DEFINITIONS FOR EVERYTHING IN THIS MANUAL

Admittance test: an electrical test (ASTM B457) of how “open” the coating still is — confirms sealing (lower = better seal).

Anode: the **positive** electrode. In anodizing it's **your part** — where the oxide grows.

Anodizing: growing an integral oxide layer out of a metal's surface by making the part the anode in an electrolyte.

Barrier layer: the thin, dense oxide at the bottom of the coating, against the metal.

Boehmite: hydrated aluminum oxide (AlOOH) formed during hot-water sealing; its swelling closes the pores.

Bright anodizing: producing a specular (mirror) aluminum surface (by bright dip or electrobrightening) and then anodizing thin and clear to protect and preserve the brilliance.

Bright dip (chemical brightening / chemical polishing): hot concentrated phosphoric-nitric (sometimes + sulfuric) acid bath (~90–110 °C) that levels and brightens the surface to a mirror — and evolves toxic NOx fumes.

Cathode: the **negative** electrode (lead or aluminum) — carries current and bubbles off hydrogen; does **not** feed the coating.

Chloride contamination: dissolved chloride that causes **pitting** (fatal to a mirror); kept very low.

Clear anodize: a thin, transparent sulfuric (Type II) anodize that lets the bright surface show through.

Constituent (intermetallic) particles: Fe/Si/Cu/Mn second-phase particles in the alloy that scatter light and **haze** a bright finish.

Current density (ASF / A/dm²): amps per unit area. Here ~12–18 ASF.

Desmut / deox: an acid dip (here usually **nitric**) that clears the film left by the bright dip.

Dimensional growth: the coating grows ~**half in, half out**; on a thin bright coat the change is small.

Distinctness-of-image (DOI): measure of how sharp a reflected image is (ASTM E430) — the truest “mirror” metric.

Drag-out / drag-in: solution clinging to a part leaving a tank (out) / contaminating the next tank (in).

Electrobrightening / electropolishing: electrochemical leveling of the surface (an alternative to chemical bright dip).

Gloss (specular): brilliance measured with a gloss meter (ASTM D523), in gloss units.

Integral coating: a coating that *is* part of the metal (grown from it) — can't peel to bare metal.

NOx (nitrogen oxides): toxic brown/orange fumes (NO, NO₂) evolved by the nitric in the bright dip; **delayed** lung injury possible — the standout hazard of this line.

Porous layer: the upper oxide full of vertical pores — why sealing (and any tint) is possible.

Purity (super-purity / 4N): 99.99% aluminum — gives the brightest finishes; bright alloys 5657/5252/5357/6463.

Sealing: the final step that **closes the pores** to lock in brilliance and corrosion resistance (without bloom).

Sealing bloom: a chalky film after sealing (hard water/wrong pH/contamination) — on a mirror it means scrap.

Significant surface: the surface that must meet gloss/thickness spec; where you measure.

Specular: mirror-like reflection (vs. diffuse/matte).

Type II: the standard sulfuric anodize — bright anodize is fundamentally a thin, clear Type II over a brightened surface.

Valve metals: metals that anodize to protective oxides (aluminum, titanium, magnesium, tantalum, niobium).

Water-break test: a free cleanliness check — clean metal sheets water; dirty metal beads it.



REMEMBER

Of all these, five matter most: **anode** (your part), **bright dip** (where the mirror — and the NOx hazard — are made), **high purity** (no mirror without it), **thin & clear anodize** (preserve the shine), and **bloom-free seal** (don't dull it at the finish line). Master those and you understand bright anodizing.

— TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO

Sign-off levels: **O** = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	Safety / PPE / SDS / hygiene — bright-dip NOx + hot acid, sulfuric, hydrogen, DC, hot seal (all stations)	—	—	—	—
2	Station 01 — Inspection & Racking (alloy/purity, contact)	—	—	—	—
3	Station 02 — Clean / Degrease (non-etching) + water-break	—	—	—	—
4	Station 03 — Bright Dip (hot phosphoric-nitric) operation + NOx/ventilation	—	—	—	—
5	Station 04 — Rinse + conductivity	—	—	—	—
6	Station 05 — Desmut / Deox (nitric)	—	—	—	—
7	Station 06 — Rinse	—	—	—	—
8	Station 07 — Clear Anodize (part = anode, thin/clear)	—	—	—	—
9	Thickness control (CD × time, run thin) & dimensional growth	—	—	—	—
10	Station 08 — Post-anodize rinse (open pores, cool DI)	—	—	—	—
11	Station 09 — Dye / light tint (or clear)	—	—	—	—
12	Station 10 — Seal (clear, bloom-free)	—	—	—	—
13	Station 11 — Rinse / Dry (spot-free)	—	—	—	—
14	Station 12 — Final Inspection (gloss/DOI, thickness, seal)	—	—	—	—
15	LOTO on rectifier / electrical safety	—	—	—	—
16	Emergency response — eyewash, shower, spill, NOx fresh-air + medical , fluoride first-aid	—	—	—	—
17	Waste segregation & environmental — phosphate/nitrate/aluminum/NOx scrubbing	—	—	—	—
18	Troubleshooting — symptom recognition (dull/hazy/orange-peel/burn)	—	—	—	—

Operator name: _____ Hire/start date: _____

Supervisor name: _____ Review cycle: ☐ Annual ☐ Other: _____



HEADS-UP

Safety rows (1, 4, 15, 16, 17) are not optional and not “fill in later.” An operator may not work *any* station — and **absolutely not the bright dip** — until PPE/SDS/hygiene, **bright-dip NOx & ventilation**, LOTO, emergency response (including **NOx fresh-air + medical evaluation** and fluoride first-aid where used), and waste-control awareness are signed. Safety competency precedes production competency.

PART FOUR — CERTIFICATION

20 QUESTIONS • PASSING SCORE 80% (16/20)

COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

**REMEMBER**

Passing score: 80% (16 of 20). Any **safety** question answered wrong (**Q3, Q9, Q13, Q17, Q19**) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: _____ Date: _____ Score: _____ / 20 **Safety-gate Qs: 3, 9, 13, 17, 19**

Q1. Anodizing differs from electroplating mainly because in anodizing the part is the:

- A. Cathode, and a different metal is deposited on it B. Anode, and an oxide is grown out of the part itself C. Insulator, and nothing happens D. Cathode, and it dissolves

Q2. The chemical “bright dip” that creates the mirror surface is typically:

- A. Cold dilute sulfuric acid B. Hot caustic (sodium hydroxide) C. Hot concentrated phosphoric-nitric acid (~90–110 °C) D. Room-temperature water

Q3. [SAFETY GATE] The standout hazard of the bright dip — beyond ordinary acid burns — is:

- A. It freezes skin B. It is harmless C. Static electricity D. Hot-acid burns PLUS toxic brown NO_x (nitrogen-oxide) fumes whose lung effects can be delayed for hours, requiring strong ventilation/scrubbing

Q4. Compared with standard satin Type II, bright anodizing deliberately:

- A. Skips sealing B. Skips anodizing C. Skips rinsing D. Skips the matte caustic etch and adds a brightening step for specular gloss

Q5. For a true mirror bright-anodized finish, the aluminum should be:

- A. High-purity / special bright alloys (e.g., 99.99%, 5657, 6463) B. A high-silicon casting alloy C. High-copper 2024 D. Any scrap aluminum

Q6. (Short answer) Explain in your own words why anodizing is **not** plating — name which electrode the part is and **where the coating comes from**.

Q7. After the bright dip, why is the anodize kept **thin and clear** rather than thick and satin?

- A. Thicker is always brighter B. To dye it black C. So the oxide is transparent and the mirror shows through, and because a thinner coating scatters less light (higher gloss) D. To hide the brilliance

Q8. The bright dip produces a mirror by:

- A. Adding a layer of metal B. Dissolving microscopic peaks faster than valleys, leveling/smoothing the surface C. Painting it bright D. Cooling it quickly

Q9. (Short answer) [SAFETY GATE] Name the **toxic gas** evolved by the phosphoric-nitric bright dip, describe its appearance, and give **two controls** that protect operators from it.

Q10. The clear anodize tank is run:

- A. Very thick and hot B. As a hot caustic C. With no current D. Thin, with the bath cold (~20–21 °C) like Type II

Q11. A bright part comes out **hazy/dull** even though the line is running fine. The most likely cause is:

- A. The wrong metal — a high-silicon/copper or low-purity alloy scatters light B. The bath is too clean C. The part is too bright D. The rectifier is off

Q12. (Short answer) List the key steps of the bright anodize line in order, and explain **why a matte caustic etch must never be added** before anodizing.

Q13. [SAFETY GATE] The seal tank (hot-water or nickel-acetate) runs at about **80–100 °C**. Its main hazard is:

A. Radiation B. None C. Scald / steam burns D. Freezing

Q14. “Dimensional growth” in anodizing means the coating grows roughly:

A. Half into the surface and half out (each surface moves out ~half the thickness) B. All inward C. All outward by 1 mm D. It shrinks the part

Q15. Coating **thickness** is controlled mainly by:

A. Dye color B. Current density × time C. Seal temperature D. The alloy only

Q16. (Short answer) Why does bright anodizing **require high-purity aluminum**? Name what iron, silicon, and copper do to the finish.

Q17. (Short answer) [SAFETY GATE] List **three** pieces of PPE required at the hot phosphoric-nitric bright dip, and state the special concern about **NOx exposure** (what makes it dangerous even when you feel okay).

Q18. Two field measurements of a bright finish's quality are:

A. pH and temperature B. Voltage and amperage C. Specular gloss (gloss meter, ASTM D523) and distinctness-of-image (DOI, ASTM E430) D. Weight only

Q19. (Short answer) [SAFETY GATE] State the universal rule for adding acid during bath make-up, and name the two main corrosive hazards on a bright line.

Q20. (Short answer) Name **two** ways bright anodizing differs from standard satin Type II anodizing.



SAFETY SECTION — MUST BE CORRECT TO CERTIFY

Questions 3, 9, 13, 17, and 19 cover hazards that can permanently harm you or others (especially the **bright-dip NOx**). A miss on any safety item means re-teach and re-test before sign-off.

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD



HEADS-UP

Keep this page out of the trainee's copy. Questions **3, 9, 13, 17, 19** are safety-critical — any wrong answer there requires re-training before sign-off, regardless of total score.

- Q1. B** — The part is the **anode**, and oxide is grown out of the part itself.
- Q2. C** — Hot concentrated **phosphoric-nitric acid** (~90–110 °C).
- Q3. D** — Hot-acid burns **plus toxic brown NOx fumes** with **delayed** lung effects; requires strong ventilation/scrubbing.
- Q4. D** — Bright anodize **skips the matte caustic etch** and adds a **brightening step** for specular gloss.
- Q5. A** — **High-purity / special bright alloys** (99.99%, 5657, 5252, 6463); castings/high-Si/high-Cu can't go bright.
- Q6.** In plating the part is the **cathode** and a **different metal is deposited onto it**. In anodizing the part is the **anode**, and the coating is **aluminum oxide grown out of the part's own surface** (oxygen from the bath + the part's aluminum) — nothing is deposited from elsewhere.
- Q7. C** — A **transparent** thin oxide lets the mirror show through, and **thinner scatters less light** (higher gloss). Thick/hazy buries the brilliance.
- Q8. B** — It **dissolves microscopic peaks faster than valleys**, leveling/smoothing the surface to a mirror (chemical leveling/polishing).
- Q9. NOx** — **nitrogen oxides** (brown/orange fumes) from the nitric acid. Controls (any two): **local exhaust / fume scrubbing, never run without ventilation**, respiratory protection per program, bath additives that suppress fuming, lids, smooth handling/short dwell. (Bonus: any exposure → fresh air + **medical evaluation** for delayed effects.)
- Q10. D** — Thin, with the bath cold (~20–21 °C) like Type II.
- Q11. A** — The **wrong metal** — high-silicon/copper or low-purity alloy scatters light and hazes. The alloy decides brilliance.
- Q12.** Order: **Inspect/Rack** → **Clean (non-etch)** → **Bright Dip** → **Rinse** → **Desmut** → **Rinse** → **Clear Anodize** → **Rinse** → (light tint, rarely) → **Seal** → **Rinse/Dry** → **Inspect**. A **matte etch must never be added** because it **frosts/mattes the surface** — the exact opposite of the mirror finish we're creating — and the damage is unrecoverable short of re-polishing.
- Q13. C** — **Scald/steam burns** (the seal runs ~80–100 °C). Face shield, heat gloves, mind the steam.
- Q14. A** — **Half in, half out** — each surface moves out ~half the coating thickness.
- Q15. B** — **Current density × time** (verify with an eddy-current gauge; run thin).
- Q16.** Bright anodize needs **high-purity aluminum** because impurities form **light-scattering constituent (intermetallic) particles**. **Iron, silicon, and copper** create **haze/frost, gray cloudiness, pitting, and dullness** — they don't anodize/dissolve uniformly with the aluminum, so they scatter light instead of reflecting it. Castings/high-Si alloys **cannot go bright**.
- Q17. PPE (any three):** sealed chemical splash goggles, face shield, heat- and acid-resistant gloves, chemical apron/suit, chemical boots. **NOx concern:** nitrogen-oxide fumes cause **deep-lung injury (pulmonary edema) that can be DELAYED for hours** — a worker may feel fine after exposure and become seriously ill later; therefore **any** exposure requires fresh air, rest, and **medical evaluation/observation** (don't wait and see), and the tank requires working ventilation/scrubbing.
- Q18. C** — **Specular gloss (ASTM D523)** and **distinctness-of-image / DOI (ASTM E430)**.
- Q19.** Universal rule: **always add ACID to WATER, never water to acid**. The two main corrosive hazards: the **hot phosphoric-nitric bright dip** and the **sulfuric anodize** (plus the **nitric desmut**) — all strong acids that cause severe burns.
- Q20.** Any two of: bright anodize **brightens (bright dip / electrobrighten) instead of matte-etching**; it requires **high-purity aluminum**; the anodize is kept **thin and clear/transparent** (vs. heavier satin); it's usually **left clear or only lightly tinted** (vs. dyed); its quality is judged by **gloss/DOI** (vs. color/coverage); the surface is **specular/mirror** (vs. satin/matte); the bright dip adds a **toxic-NOx-fume hazard** the satin line doesn't have.
- Answer-key letter distribution (MC):** A: Q5, Q11, Q14 · B: Q1, Q8, Q15 · C: Q2, Q7, Q13, Q18 · D: Q3, Q4, Q10 — a healthy spread across all four letters (A 3 · B 3 · C 4 · D 3).



REMEMBER

A score of 16/20 passes — but every safety question (3, 9, 13, 17, 19) must be correct to certify. Miss a safety item → re-teach and re-test before sign-off. On a bright line, the bright-dip NOx hazard is the one we never gamble on.

— PRINT, FILL IN, SIGN



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This certifies that

OPERATOR NAME

has successfully completed the **Bright Anodizing (Mirror / Specular Decorative Finish)** training course — covering the full bright-anodize workflow (Inspection & Racking, Non-Etching Clean, **Chemical Bright Dip**, Rinse, Desmut/Deox, **Clear Anodize**, Seal, Rinse/Dry, and Final Inspection), the fundamental concept that **the part is the anode and the oxide is grown out of the part itself**, the **brighten-then-clear-anodize** principle, the requirement for **high-purity aluminum**, the porous-oxide / seal principle, dimensional growth, thin-clear coating control by current density and time, electrical-contact and racking practice, gloss/DOI measurement, and — above all — the **safety practices** required of every operator (the **hot phosphoric-nitric bright dip and its toxic, delayed-effect NOx fumes**, sulfuric-acid burn control, hot-seal scald control, hydrogen and DC-electrical controls, fluoride first-aid where applicable, spill response, ventilation/scrubbing, and waste segregation) — and has passed the Completion Test with a score of _____ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: _____ Re-certification due: _____

TRAINEE SIGNATURE

CERTIFYING TRAINER

SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS (bright-dip baths are typically proprietary), customer/architectural/automotive specifications (e.g., ASTM B580; MIL-PRF-8625 Type II for the clear anodize), and applicable regulatory requirements before production use. The hot phosphoric-nitric bright dip causes severe burns and evolves toxic NOx fumes; sulfuric acid causes severe burns; seal tanks run near boiling — always consult current SDS and comply with OSHA, EPA, REACH, and local regulations.

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